GENERAL SCIENCE GRADE 10: INTRODUCTION TO GENERAL SCIENCE

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 1.1 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	General Science
Strand	Strand 1.0: Life Science
Sub-Strand	Sub-Strand 1.1: Introduction to General Science
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– The meaning and importance of General Science as a learning area – Importance of General Science in human life, environment and technology – Careers in General Science – The principle of inference as used in science education
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) explain the meaning and importance of General Science as a learning area, b) outline the importance of General Science in life, environment and technology, c) identify career opportunities related to General Science, d) analyse the principle of inference in science education, e) appreciate the importance of General Science in day-to-day life.
Core Competencies	– Self-efficacy: as the learner designs and presents charts on career opportunities related to General Science and identifies a particular career path of own interest. – Digital literacy: as the learner uses digital media to research on the scientific processes, importance of General Science and careers related to General Science. – Citizenship: as the learner draws a chart on careers related to General Science and identifies with a particular career path for individual and national development.
Core Values	– Respect: as the learner appreciates others' opinions when discussing the importance of General Science in day-to-day life. – Responsibility: as the learner uses digital devices appropriately to search for information on the career opportunities related to General Science.
Science & Engineering Practices	– Asking Questions and Defining Problems: as the learner brainstorms and formulates questions about what General Science is and why it matters, anchored in the puzzling phenomenon of how scientists make conclusions from indirect evidence. – Obtaining, Evaluating, and Communicating Information: as the learner uses digital and print media to research scientific processes, the importance of General Science, and related careers, then shares findings with peers. – Analysing and Interpreting Data: as the learner explores the process of collecting evidence and drawing inferences in science education, analysing how scientists move from raw data to conclusions. – Constructing Explanations: as the learner applies the principle of inference to explain observable phenomena in everyday

	Kenyan contexts (e.g., inferring soil health from crop yield in Rift Valley farms).
Pertinent & Contemporary Issues (PCIs)	– Socio-Economic and Environmental Issues: as the learner discusses with peers the importance of General Science to the environment and technology.
Career Connections	– Medical doctor / Clinical officer: General Science provides the biological and chemical foundation for diagnosing and treating disease in Kenyan hospitals and health centres. – Agricultural scientist / Agronomist: understanding life science and scientific processes supports improved crop production and food security across Kenya's farming regions such as Kericho and the Rift Valley. – Environmental scientist: knowledge of General Science underpins environmental monitoring and conservation of ecosystems like Lake Victoria and the Mau Forest. – Laboratory technologist: scientific processes and inference skills are directly applied in government and private laboratories across Nairobi, Kisumu and Mombasa. – Science teacher / Educator: mastery of General Science content and pedagogy enables learners to inspire the next generation of Kenyan scientists. – Biotechnology researcher: General Science builds the conceptual foundation for innovation in areas such as seed technology, vaccine development and biogas production relevant to Kenya's Big Four Agenda.
Focus for Lessons	This unit launches learners into Senior School General Science by grounding them in what science is, why it matters in Kenyan society, and how scientists think. Using an anchoring phenomenon drawn from everyday Kenyan life — a detective-style mystery about how a farmer in Kisumu drew conclusions about a pest infestation without ever seeing the pest directly — learners explore the nature of scientific inference, the breadth of career pathways opened by General Science, and the relevance of the discipline to health, food, environment, and technology in Kenya. The unit foregrounds Science and Engineering Practices from the outset, particularly asking questions, obtaining and evaluating information, and constructing explanations from evidence, while building the classroom norms of collaborative sensemaking that will sustain the full year of inquiry.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How does a scientist know something is true when they cannot see it directly — and how is this power of General Science shaping life, careers, and the environment in Kenya? KICD KEY INQUIRY QUESTION: 1. How is General Science useful in daily life?
Anchoring Phenomenon	A maize farmer in Kisumu notices that her maize plants along one edge of the shamba are wilting and yellowing, even though she has watered and fertilised them the same as the rest of the field. She has never seen any insect on those plants, yet she concludes — correctly — that a soil pest is destroying the roots underground. Students encounter this scenario: How did she know? She never observed the pest directly. This is puzzling because it challenges the common assumption that "seeing is believing" in science. The farmer used indirect evidence — the pattern of wilting, the location, the time of season, the absence of other causes — to infer a hidden cause. This phenomenon anchors the entire sub-strand: What is General Science, and how do scientists make reliable conclusions from evidence they cannot directly observe? It naturally threads into the importance of science in agriculture and food security (a deeply Kenyan concern), the range of scientific careers that depend on inference (doctors reading X-rays, lab technologists testing blood, environmental scientists measuring pollution), and the formal principle of inference as a core scientific practice.
Storyline Thread	Lesson 1 — Anchoring Phenomenon & Initial Predictions (Predict Phase): Teacher presents the Kisumu maize-farmer scenario on the board with a photograph of a wilting maize field. Students individually write their best explanation for why the border plants are dying — no instruction yet, pure prior knowledge activation. Cold-call 4–5 students (wait time: 10 seconds each). Teacher records all predictions on a class anchor chart titled "What We Think We Know." Students are introduced to the Driving Question Board

	(DQB) — a large poster where they post sticky-note questions they cannot yet answer. Lesson closes with teacher posing: "If the farmer never saw the pest, how could she possibly know it was there?" Lesson 2 — What is General Science? Mapping the Discipline (Observe Phase – Research): Students use available digital devices or print media (textbooks, magazines, resource persons) to brainstorm and research the meaning of General Science as a learning area. In pairs, learners identify at least three ways General Science connects to life in their own community (food, health, environment, technology). Groups present a 2-minute summary. Teacher facilitates sensemaking: General Science is not just facts — it is a way of asking questions and finding answers. The Kisumu farmer is doing science. Class revisits DQB and adds new questions. Lesson 3 — Importance of General Science in Life, Environment and Technology (Observe Phase – Discussion & Media): Teacher guides learners to discuss with peers the importance of General Science in relation to human life (nutrition, disease), environment (conservation, pollution), and technology (mobile phones, water treatment). Learners use print or digital media to find one local Kenyan example for each category — e.g., KEBS food safety testing, NEMA environmental monitoring, M-PESA technology built on engineering science. Each group presents one example; teacher connects every example back to the anchoring phenomenon: "In each case, someone had to infer something they could not directly see." Lesson 4 — Careers in General Science (Observe Phase – Career Research & Chart Making): Learners use digital devices or print media to research career opportunities related to General Science. Each learner draws a personal career-path chart linking General Science sub-topics (biology, chemistry, physics) to specific Kenyan careers (doctor, agronomist, lab tech, environmental scientist, teacher, engineer). Charts are displayed on the classroom wall. Learners identify with a particular career path and write a 3-sentence justification. Teacher cold-calls 5 learners to share their choice and reasoning (wait time: 8 seconds each). DQB updated: "What science skills do I need for my chosen career?" Lesson 5 — Introduction to Scientific Processes and the Principle of Inference (Explain Phase – Part 1): Teacher introduces the principle of inference formally: inference = a conclusion drawn from evidence + reasoning, not from direct observation. Learners revisit the Kisumu farmer scenario and now formally identify: (a) the evidence she had, (b) the reasoning she applied, (c) the inference she made. Class works through two additional supporting phenomena (Nairobi doctor / Amboseli ranger) using the same evidence-reasoning-inference framework. Learners record a structured table in their exercise books. Teacher emphasises: this is the foundation of all scientific knowledge. Lesson 6 — Collecting Evidence and Drawing Inferences: Hands-On Activity (Explain Phase – Part 2): Learners carry out a guided classroom activity: Teacher places 4–5 mystery "evidence bags" (sealed envelopes containing written clues — e.g., descriptions of a plant, soil colour, smell, leaf texture — no images). Groups must infer what is inside the bag (a plant disease? a nutrient deficiency? a pest?) using only the written evidence. Groups present their inferences and justify with evidence. Teacher leads discussion: What methods do scientists use to collect evidence? (observation, measurement, experimentation, record-keeping.) Class maps these onto the scientific process skills named in the KICD Essence Statement: classifying, measuring, hypothesising, identifying variables, interpreting data. Lesson 7 — DQB Curation and Model Building (DQB Creation & Model Building Phase): Class returns to the DQB and categorises questions into three columns: "We can now answer this," "We partly understand this," "We still don't know this." For questions now answerable, learners collaboratively write 1–2 sentence explanations using evidence from lessons 1–6. For remaining questions, teacher previews how future sub-strands (The Cell, Nutrition, Respiration, etc.) will provide answers — connecting the unit storyline to the full course. Each learner then creates or updates their personal Cumulative Science Model: a mind-map or diagram showing how General Science connects to their chosen career, to the anchoring phenomenon, and to at least two real Kenyan contexts.

	Pairs peer-review each other's models using the KICD assessment rubric descriptors. Lesson 8 — Synthesis, Presentations and Unit Assessment (Model Building Phase – Consolidation & Assessment): Learner groups present their career charts and inference-framework summaries to the class (2 minutes per group). Teacher uses structured cold-calling (every group must contribute; wait time 10 seconds) to elicit connections: "How does the principle of inference appear in your chosen career?" Class co-constructs a final anchor-chart answer to the Driving Question. Teacher administers a short written formative task: given a new Kenyan scenario (e.g., a tea farmer in Kericho noticing unusual leaf curl), learners must (a) identify evidence, (b) state a reasonable inference, and (c) name one General Science career that would investigate this professionally. Unit closes with teacher previewing Sub-Strand 1.2 (The Cell): "To understand how the maize root is being destroyed, we first need to understand what a cell is — that is where we go next."
Supporting Phenomena	– A Nairobi paediatrician diagnoses malaria in a child without seeing the Plasmodium parasite directly — she infers it from fever patterns, blood smear results, and symptoms, illustrating inference in medical science. – A Kenya Meteorological Department forecaster at Wilson Airport predicts rain for Kisumu 48 hours in advance without being able to "see" tomorrow's weather — learners explore how scientists infer future events from current data patterns. – A food scientist at a Kenyan uji (porridge) processing factory in Eldoret detects aflatoxin contamination in maize flour using a UV lamp, even though the toxin is invisible to the naked eye — a striking example of how General Science tools extend human inference beyond direct observation. – A wildlife ranger in Amboseli National Park determines that lions passed through an area the previous night by reading paw prints, disturbed vegetation, and prey remains — a familiar Kenyan context showing that inference is not unique to laboratories but is a universal human-science skill.

LESSON 1 (40 minutes): How Did She Know? — Anchoring the Phenomenon & Opening the Driving Question Board	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To launch the sub-strand by anchoring learners in a puzzling, Kenya-relevant phenomenon that reveals how scientists — and ordinary Kenyans — draw reliable conclusions from evidence they cannot directly see, and to open the Driving Question Board that will track growing understanding across the sub-strand.
Knowledge	Learners will explain the meaning of General Science as a learning area and begin to outline its importance in daily life, environment and technology in Kenya.
Skills	Learners will practise the NGSS Science and Engineering Practice of asking questions and defining problems; they will record prior-knowledge predictions in writing, share reasoning orally, and post initial questions on the Driving Question Board.
Attitudes	Learners will appreciate that curiosity and careful observation — even of things we cannot see directly — are the foundations of General Science, and will begin to respect others' ideas during whole-class discussion.
Key Inquiry Question	How is General Science useful in daily life?
Purpose in Storyline	This is the opening lesson of the sub-strand. It presents the anchoring phenomenon — a maize farmer in Kisumu who correctly identifies a hidden soil pest from indirect evidence alone — and uses it to surface learners' prior knowledge, reveal misconceptions, and generate the questions that will drive the entire sub-strand. No content is taught yet; the lesson exists purely to create intellectual need and to establish the Driving Question Board.
Safety Notes	No laboratory activity in this lesson. Ensure all learners feel psychologically safe to share predictions without fear of being 'wrong' — explicitly state before cold-calling: 'There are no wrong answers yet; we are scientists collecting all possible ideas.'

B. LESSON OVERVIEW
Learners encounter a photograph of a wilting maize field on the border of a shamba near Kisumu. The teacher presents the farmer's story: she watered and fertilised all plants equally, never saw any insect, yet concluded — correctly — that a soil pest was destroying the roots underground. Without any instruction, learners individually write their best explanation, share predictions in a cold-call discussion, and see all ideas recorded on an anchor chart titled 'What We Think We Know.' The lesson closes with the launch of the Driving Question Board (DQB), where learners post sticky-note questions they cannot yet answer, and the teacher poses the sub-strand's closing hook: 'If the farmer never saw the pest, how could she possibly know it was there?' This lesson activates prior knowledge, reveals misconceptions, and establishes the intellectual need that motivates the entire sub-strand.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** When to use z or t statistics in	Each learner silently reads the scenario printed on the board or projected on screen: 'A maize farmer in Kisumu notices that the	1. Display the photograph of a wilting maize field (border plants yellow, interior plants green) on the board or projector. Say exactly:	Individual Written Prediction (Think Before You Talk). This strategy ensures every learner commits to a position before social pressure	Teacher scans written predictions during circulation. Observable indicators: (a) Does the learner attempt a causal explanation (e.g.,

significance tests Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Scale and Indirect Measurement Applications (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	plants along one edge of her shamba are wilting and turning yellow. She has watered and fertilised every plant exactly the same way. She has never seen any insect on those plants. Yet she tells her neighbour with confidence: there is a pest eating the roots underground. How does she know?' Learners individually write their best explanation in their exercise books under the heading: 'My Prediction — What I Think Is Happening.' They must write at least two full sentences and give a reason for their thinking. No talking during this phase.	'Look at this photograph. This is a real shamba near Kisumu. Take thirty seconds — just look. What do you notice?' (Wait 30 seconds in silence.) 2. Read the scenario aloud slowly, pointing to the photograph at each detail. 3. Say: 'Now, in your exercise books, write your best explanation. You have four minutes. Write at least two sentences and give a reason. Do not talk to anyone yet — I want YOUR thinking, not your neighbour's.' 4. Circulate silently for 4 minutes, scanning books without commenting. Note 2–3 interesting or divergent predictions to use in the cold-call round. Do NOT correct anyone. 5. After 4 minutes say: 'Pens down. Now we are going to hear from some of you.'	shapes their thinking. It surfaces the full range of prior knowledge and misconceptions in the room, giving the teacher diagnostic data and giving learners a personal stake in the phenomenon they will later explain with science.	'something in the soil') or only describe what they see? (b) Does any learner spontaneously use the word 'evidence,' 'infer,' or 'signs'? (c) Are predictions based on the visible pattern (border location, uniform watering) or purely on guessing? Teacher mentally flags 4–5 learners representing a range of explanation quality for cold-calling in the next phase.
Observe Phase 📹 **VIDEO:** When to use z or t statistics in significance tests Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Scale and Indirect Measurement Applications (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners share their predictions in a cold-call round. They listen to classmates' ideas and observe the teacher recording every prediction — without judgment — on a large anchor chart titled 'What We Think We Know' displayed at the front of the class. Learners notice that different people reached different conclusions from the same scenario, and that some ideas contradict each other. They are not yet asked to evaluate or rank the ideas.	1. Point to the blank anchor chart. Say: 'I am going to write down everything you say on this chart. Every idea goes up — I am not going to tell you if you are right or wrong yet. We are collecting all our thinking.' 2. Cold-call exactly 5 learners (do not ask for volunteers first — choose the 4–5 flagged during circulation plus one more). Use a rotating name strategy so learners know anyone can be called. For each learner: state their name, pause (WAIT TIME: 10 seconds — count silently), then invite: '[Name], what did you write? Tell us your explanation.' If a learner says 'I don't know,' say: 'Read me what you wrote in your book' — this ensures no one is left silent. 3. After each response, paraphrase and write it on the chart in the learner's own words. Do NOT evaluate. Say only: 'Thank you — that idea is now on	Anchor Chart — 'What We Think We Know.' This is a whole-class knowledge inventory. Making predictions public creates accountability (learners are invested in resolving their own ideas) and models the scientific practice that all hypotheses must be stated before evidence is gathered. It also allows the teacher to diagnose the spread of prior knowledge and target instruction accordingly.	Observable indicators: (a) At least 4 of 5 cold-called learners can state a causal reason (not just describe the symptom). (b) At least one learner references location pattern ('the border') as a clue — this indicates readiness to think about indirect evidence. (c) Teacher notes any learner who spontaneously uses scientific vocabulary (evidence, cause, inference, pattern) — these learners can serve as peer anchors in later lessons. Record on a class observation checklist.

		our chart.' 4. After all 5 responses, ask the whole class: 'Hands up if you wrote something different from everything on this chart.' Cold-call 2 more learners from those who raise hands. Add to chart. 5. Draw a box around the chart and say: 'This is what WE — as a class — think we know right now. We will come back to this chart at the end of the sub-strand to see what changed.'		
Explain Phase 📹 **VIDEO:** When to use z or t statistics in significance tests Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Scale and Indirect Measurement Applications (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher does NOT explain the answer to the phenomenon yet. Instead, learners are guided through a short structured discussion that sharpens the puzzle: What kinds of information did the farmer actually use? Learners work in pairs for 3 minutes to list every piece of information the farmer could have had — without seeing the pest. Pairs share out. The teacher then briefly introduces the idea that General Science is precisely the discipline that deals with this challenge: drawing reliable conclusions from evidence, including evidence we cannot see directly. The class is introduced to the names of two real career paths — a soil scientist (pedologist) at the Kenya Agricultural and Livestock Research Organisation (KALRO) and a medical lab technologist at Kenyatta National Hospital — as people who make life-changing decisions from indirect evidence every day.	1. Say: 'Before I tell you whether any of our predictions are correct — let us think more carefully. Turn to the person next to you. You have three minutes. List every piece of information the farmer could have noticed WITHOUT picking up a single insect. Go.' (Set a 3-minute timer visibly.) 2. Circulate; listen for: wilting pattern is on the border only; the rest of the field is healthy; she watered and fertilised equally; the time of year; the colour of leaves; the soil texture. Do not prompt yet — just listen. 3. After 3 minutes: cold-call 3 pairs (WAIT TIME: 10 seconds each). Write their evidence list on the board in a second column next to the anchor chart. Label this column: 'Evidence the Farmer Had.' 4. Say: 'Look at that list. She never saw the pest. But she had ALL of that. Scientists have a word for what she did — she made an INFERENCE. That word will become very important in this sub-strand.' Write INFERENCE on the board. Do not define it fully yet — leave it as a teaser. 5. Say: 'A scientist at KALRO in Nairobi does exactly this when she analyses soil samples. A lab technologist at Kenyatta National Hospital does exactly this when she reads a	Think-Pair-Share + Evidence Sorting. By asking learners to list evidence before they know the answer, the teacher builds the cognitive habit of separating observation from conclusion — a foundational scientific practice. Connecting the farmer's reasoning to KALRO scientists and KNH lab technologists grounds abstract scientific thinking in tangible Kenyan careers, making the purpose of General Science personally meaningful.	Observable indicators: (a) In the pair discussion, are learners listing observable clues (location, colour, pattern) rather than invented causes? (b) During cold-call share-out, can at least 2 pairs distinguish between 'what the farmer saw' and 'what she concluded'? This distinction is the conceptual seed of inference — its presence or absence tells the teacher how much scaffolding the principle-of-inference lessons will need. Record names of learners who make this distinction spontaneously.

		blood test. They never see the disease directly — they read the evidence. That is General Science at work in Kenya, today.' 6. Say: 'We are not going to answer the full question today. That is the point. Today we are opening the investigation.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [When to use z or t statistics in significance tests]() Source: Khan Academy (English - US curriculum) [🔍 Search ARES for similar videos]() 📖 **HTML:** [Scale and Indirect Measurement Applications (Flexbook)]() Source: CK-12 [🔍 Search ARES for similar readings]()	Each learner receives two sticky notes. On the first, they write one question about the farmer's situation or about General Science that they genuinely cannot answer yet. On the second, they write one question about a career or a real-life situation in Kenya where they think this kind of 'hidden evidence' science matters. Learners walk to the front and post their sticky notes on the class DQB — a large poster divided into three columns: 'Questions about the Phenomenon,' 'Questions about General Science,' and 'Questions about Careers & Kenya.' The teacher reads 4–5 posted questions aloud, validates them, and explains that the DQB will grow and be updated every lesson until all questions are answered.	1. Distribute two sticky notes per learner. Say: 'You have two minutes. On your FIRST sticky note, write one question you have about the farmer's situation — something you genuinely do not know yet. On your SECOND sticky note, write one question about science or careers in Kenya where this kind of hidden-evidence thinking is used. Your name goes in the corner of each note.' 2. Give exactly 2 minutes (timer visible). 3. Say: 'Come up row by row and place your notes in the correct column on the board.' Supervise posting — redirect any notes placed in the wrong column with: 'That sounds more like a question about [X] — can you move it to that column?' 4. Once all notes are posted, read 4–5 aloud — choose at least one that connects to careers, one that connects to the phenomenon, and one that explicitly asks 'How do we know something we cannot see?' Say after each: 'That is a great scientific question. We will answer that in this sub-strand.' 5. Say: 'This board belongs to all of us. Every lesson, we will add questions and — when we have earned the answer — move questions to the ANSWERED column. Treat it like a living scientific notebook for our class.' 6. Photograph the DQB for the ARES platform record.	Driving Question Board (DQB) — Question Generation Protocol. The DQB externalises curiosity and makes the learning goals student-owned rather than teacher-imposed. By categorising questions publicly, learners see the shape of the sub-strand's investigation space. Questions posted on Day 1 will be revisited in every subsequent lesson, creating a continuous narrative of growing understanding — the hallmark of phenomenon-based learning.	Observable indicators: (a) Are learners' questions about the phenomenon or are they purely factual recall questions ('What is science?')? Genuine inquiry questions ('Why would only the border plants be affected?') signal higher-order thinking. (b) Do any questions reference specific Kenyan contexts (KEBS testing, soil scientists, doctors reading X-rays at Nairobi Hospital) — indicating transfer of the explanation-phase discussion? (c) Count: at minimum 70% of sticky notes should contain a genuine question (a sentence with a question mark and a knowledge gap), not a statement. Teacher reviews all notes after class and uses them to plan targeted instruction for Lessons 2–4.

Model Building Phase 📹 **VIDEO:** When to use z or t statistics in significance tests Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Scale and Indirect Measurement Applications (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	In the final 5 minutes, each learner draws a quick initial model in their exercise book. The prompt is: 'Draw a simple diagram showing how the Kisumu farmer moved from what she SAW to what she CONCLUDED. Use arrows. Label what is evidence and what is the conclusion.' This is a first-draft model — rough and incomplete is expected and acceptable. Learners do not share this model today; it is sealed in their books as a baseline that they will revise in every subsequent lesson as they learn more about the principle of inference, scientific processes, and General Science careers.	1. Say: 'In the last five minutes, I want you to draw — not write — a quick diagram in your book. Show me: how did the farmer get from what she observed to what she concluded? Use arrows. Label your diagram with the words EVIDENCE and CONCLUSION. You do not need to be an artist. A rough diagram with labels is perfect.' (Write the words EVIDENCE and CONCLUSION on the board as scaffolds.) 2. Give exactly 4 minutes (timer visible). Circulate — do not comment on content, but prompt any learner who stares blankly: 'Just draw the maize plant, draw a question mark underground, and draw an arrow to the farmer's conclusion — start there.' 3. At 1 minute remaining, say: 'Finish the sentence at the bottom of your diagram: The farmer knew because she ____________.' 4. Say: 'Close your books. We are NOT sharing these today. This is your starting point. By the end of this sub-strand, you will come back to this page and you will be amazed at how much your model has grown.' 5. Close with the driving hook — say slowly and deliberately: 'Here is the question that will follow us for the next several lessons: If the farmer never saw the pest — how could she possibly know it was there? Think about that tonight.' Write on the board: DRIVING QUESTION: How does a scientist know something is true when they cannot see it directly?	Initial Model Construction (Baseline Diagram). Drawing a model forces learners to make their mental representation of causality visible and concrete. An arrow-based diagram of EVIDENCE → CONCLUSION is the simplest possible representation of the inference process — and it is revisable. Each subsequent lesson will add complexity to this model (adding variables, mechanisms, named scientific processes), creating a cumulative record of conceptual growth across the sub-strand.	Observable indicators: (a) Does the learner's diagram include at least one arrow connecting an observation to a conclusion? (b) Does the learner label any element as 'evidence' — even loosely? (c) Does the closing sentence ('The farmer knew because she ___') reference something observable rather than simply stating 'because she is clever'? Teacher collects a sample of 6–8 books at the end of class (varied ability range) to photograph the baseline models for comparison at the sub-strand's end. These also inform differentiation in Lesson 2.

D. TEACHER REFLECTION

After the lesson, ask yourself: (1) Did the cold-call round surface at least two genuinely different types of explanations — one focused on the visible symptoms and one

that attempted to name a hidden cause? If all 5 predictions were identical, the phenomenon may not have created enough cognitive conflict; consider introducing a contrasting case in Lesson 2 (e.g., a farmer who does NOT use indirect evidence and makes the wrong call). (2) Were the DQB questions mostly superficial ('What is science?') or genuinely inquiry-based ('How can you prove something exists if you cannot see it?')? If superficial, spend 5 minutes at the start of Lesson 2 modelling what a strong scientific question looks like using one of the better examples from today's board. (3) Did learners' initial models show an arrow from evidence to conclusion, or did they draw the pest directly without showing the inference step? If most models skipped the inference step, the principle-of-inference lessons (Lessons 3–4) will need heavier scaffolding with worked examples from Kenyan contexts (e.g., a doctor at KNH inferring malaria from a blood smear; a KEBS inspector inferring food spoilage from pH data). (4) Were all learners comfortable during cold-calling, or did the 10-second wait time feel uncomfortable for some? If anxiety was visible, introduce a 'phone a friend' option in Lesson 2 — the cold-called learner may ask one classmate for support before answering. (5) Note which learners spontaneously used the words 'evidence,' 'pattern,' or 'infer' — these learners can be strategic pair partners for less confident peers in the Observe and Explain phases of Lessons 2 and 3.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	A photograph of a wilting maize field near Kisumu: the border plants are yellow and drooping while the interior plants are green and healthy, despite identical watering and fertilisation across the entire shamba.
What did I learn?	Scientists — and observant Kenyan farmers — can draw reliable conclusions about things they cannot see directly by carefully reading patterns of visible evidence. This practice is called inference, and it is one of the most powerful tools in General Science. It is used daily by Kenyan professionals: soil scientists at KALRO, lab technologists at Kenyatta National Hospital, and environmental scientists monitoring Lake Victoria.
How does this explain the phenomenon?	The farmer inferred a hidden soil pest because the pattern of wilting (border only, not random), the elimination of other causes (equal water and fertiliser), the time of season, and her knowledge of local pests all pointed to one hidden explanation. She did not need to see the pest — the evidence spoke for it. This is the opening puzzle of our sub-strand: How does General Science allow us to know what we cannot directly observe?

LESSON 2 (40 minutes): What is General Science? Mapping the Discipline	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students research the meaning of General Science as a learning area and map its connections to community life in Kenya, building the conceptual foundation needed to explain how the Kisumu farmer's reasoning counts as scientific thinking.
Knowledge	Explain the meaning of General Science as a learning area and outline its importance in life, environment and technology.
Skills	Use available digital and/or print media to research scientific processes; communicate findings in a structured 2-minute group summary.
Attitudes	Appreciate the importance of General Science in day-to-day life and respect peers' opinions when discussing its relevance to the community.
Key Inquiry Question	How is General Science useful in daily life?
Purpose in Storyline	In Lesson 1 students encountered the puzzle of the Kisumu farmer who inferred a hidden soil pest from indirect evidence. This lesson answers the question that puzzle raises: what kind of thinking is she actually doing? By mapping General Science as a discipline — its meaning, scope, and community relevance — learners establish that the farmer is practising science, setting up the deeper investigation of inference (Lessons 3–5) and careers (Lesson 6).
Safety Notes	No laboratory hazards. If learners use personal digital devices, remind them of responsible digital citizenship: use school-approved search terms, do not share personal data, and report any inappropriate content to the teacher immediately.

B. LESSON OVERVIEW
Learners work in pairs to brainstorm and then research the meaning of General Science using available digital devices or print media (textbooks, magazines, resource persons). Each pair identifies at least three ways General Science connects to life in their own community — food, health, environment, or technology — using Kenyan examples. Pairs combine into groups of four to prepare a 2-minute summary. Groups present to the class. The teacher facilitates whole-class sensemaking around the idea that General Science is not merely a collection of facts but a systematic way of asking questions and finding answers — and that the Kisumu maize farmer is doing exactly that. The class revisits the Driving Question Board (DQB) and adds new questions generated by today's research.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Scale and Indirect Measurement Source: CK-12 🔍 Search ARES for similar videos	Working individually (2 min silent think), each learner writes two responses in their exercise book: (a) 'In one sentence, what do YOU think General Science is?' and (b) 'Give one example from your village, town, or home where someone is doing something that	1. Display the prompt on the board or read it aloud twice: 'What is General Science? Write your own definition and give one real example from your community.' 2. Say: 'You have two minutes — write in your own words, there is no wrong answer right now.' Set a	Think-Write-Pair (individual prediction before instruction): activates prior knowledge and creates a personal stake in the research that follows. By committing a prediction to paper, learners have a concrete claim to confirm, revise, or reject —	Circulate during silent writing and scan exercise books. Look for: (a) whether learners can produce any working definition, however informal; (b) whether community examples are locally grounded (e.g., a mama mboga judging ripeness, a shamba boy mixing

📖 **HTML:** Using the General Absolute Value Equation and the Graphing Calculator (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	looks like science — even if they don't call it that.' Learners then share with an elbow partner (1 min). No whole-class sharing yet — predictions are held privately to be tested against research findings.	visible timer. 3. After 2 min, say: 'Share your idea with the person next to you. Listen carefully — you will use their idea later.' Allow 1 min. 4. Do NOT take whole-class answers yet. Say: 'Hold your ideas — let us go and find some evidence to test them.'	mirroring the scientific practice of forming a hypothesis before gathering evidence.	fertiliser, a nurse taking temperature). Flag learners who leave the page blank — seat them next to a more confident partner during the research phase.
Observe Phase 📹 **VIDEO:** Scale and Indirect Measurement Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Using the General Absolute Value Equation and the Graphing Calculator (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Pairs use available digital devices (phones, tablets, school computers) or print resources (Grade 10 General Science textbook, magazines, KICD curriculum card) to research: (a) the definition of General Science, (b) at least three areas of life in Kenya where General Science is important (choosing from: food and agriculture, health and medicine, environment, technology and industry). Each pair records findings in a simple two-column table in their exercise book: 'What General Science is' | 'How it connects to our community'. After 8 minutes of pair research, pairs combine into groups of four and spend 4 minutes preparing a 2-minute spoken summary. One group member is the timekeeper, one is the speaker, one records the group's three community connections on a half-sheet of paper to be posted on the class board.	1. Distribute or confirm access to resources. Say: 'You have eight minutes in pairs. Your mission: find out what General Science means and write down at least three ways it touches life here in Kenya — think about food, health, environment, technology. Use your textbook, a device, or ask me.' 2. Circulate actively. After 3 min, cold-call two pairs: 'Amina and Brian, what is the most interesting connection you have found so far?' (Wait 10 seconds for response.) 'Joseph and Faith, do you agree or have a different one?' 3. After 8 min, say: 'Pairs, join another pair — you are now a group of four. In four minutes, agree on your three best community connections and choose one person to share them with the class. You will have exactly two minutes to present.' 4. While groups prepare, write on the board: 'General Science is… / It matters in Kenya because… / One thing that surprised us…' — this is the scaffold for presentations.	Collaborative Jigsaw Research with Structured Reporting: each pair acts as a mini-research team, then pools findings with another pair. The structured reporting scaffold ('General Science is… / It matters because… / One surprise…') forces learners to synthesise rather than just copy, and the 2-minute time limit requires prioritisation — both are core science-communication skills.	During circulation, listen for whether pairs are connecting General Science to concrete Kenyan contexts (e.g., soil science helping maize farmers in the Rift Valley, medical lab technologists testing blood in Kenyatta National Hospital, NEMA scientists measuring pollution in Nairobi rivers). If pairs are producing only abstract definitions copied from text, prompt: 'That is a good start — now, where do you see this in Kisumu or your own home area?' Record which groups have strong local connections vs. generic definitions — use this to sequence the presentations (strong examples last, to build to the sensemaking phase).
Explain Phase 📹 **VIDEO:** Scale and Indirect Measurement Source: CK-12 🔍 Search ARES for similar videos	Groups present their 2-minute summaries one at a time while the rest of the class listens and ticks off any community connection they also found ('Yes, we had that too') or marks a star if it is a new connection they did not think of.	1. Call groups to present in sequence (weakest-to-strongest community connections based on your circulation notes). For each presentation, ask ONE follow-up question — choose from: 'How does this connect to food security	Predict–Observe–Explain Annotation (POE Annotation): learners return to their written prediction and annotate it using a three-action protocol (confirm / revise / add). This makes conceptual change visible to both	Collect the half-sheets posted on the board (three community connections per group). Scan for: (a) diversity of connections across food, health, environment, technology; (b) genuine Kenyan specificity (names of places, crops,

📖 **HTML:** [Using the General Absolute Value Equation and the Graphing Calculator (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	After all groups present, learners individually revisit their opening prediction in their exercise books and annotate it: underlining anything that was confirmed, crossing out anything they now disagree with, and adding at least one new idea in a different colour. The class then co-constructs a shared one-sentence definition on the board. Teacher uses the Kisumu farmer story to clinch the concept: the farmer is doing General Science.	in Kenya?' / 'Could a farmer in Kericho use this?' / 'Is the farmer in our phenomenon doing this kind of science?' 2. After all presentations, say: 'Look back at what you wrote at the start of the lesson. Underline what turned out to be true. Cross out what you now think is wrong. Add one new idea you did not have before — use a different colour pen or pencil.' Allow 2 min. 3. Cold-call three learners (choose one who had a thin prediction, one who had a strong prediction, one who is usually quiet): 'Wanjiru, what did you add to your prediction?' (Wait 12 seconds.) 'Otieno, what did you cross out and why?' (Wait 12 seconds.) 'Aisha, did anything surprise you?' (Wait 10 seconds.) 4. Facilitate co-construction: 'Using everything we have heard, who can give me a sentence that starts: General Science is…?' Accept two or three offers, then guide the class to agree on one sentence and write it prominently on the board. 5. Return to the phenomenon: 'The maize farmer in Kisumu — she watered her crop, she fertilised it, she observed the wilting pattern, she reasoned about the cause she could not see. Is she doing General Science?' Pause 15 seconds. Take three cold-calls. Affirm: 'Yes — General Science is a way of asking questions and finding answers using evidence. She is doing exactly that.'	teacher and learner, and directly mirrors the scientific practice of revising claims in light of new evidence — a meta-cognitive move that prepares learners for the inference lessons ahead. The co-constructed definition further consolidates shared understanding.	careers). During the annotation step, circulate and look for learners who are only confirming (not revising or adding) — this signals over-confidence or shallow reading; prompt them: 'Did any group mention something you had not thought of?' Exit-check: ask two or three learners at random: 'In one breath — what is General Science?' They should be able to produce a working sentence.
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Scale and Indirect]	Each learner receives one sticky note (or tears a small piece of paper). They write ONE new question that today's research has raised for them — something they now want to know more about. Learners post their sticky notes on	1. Direct learners to the DQB (a section of the board or a chart). Say: 'You now know more about General Science than you did 30 minutes ago. But knowing more often means asking better questions. On your sticky note,	Driving Question Board Question Generation: learners generate questions rather than just answering them, practising the NGSS science practice of 'Asking Questions and Defining Problems'. Publicly posting questions makes	Read all posted sticky-note questions before the next lesson. Sort them into: (a) questions already answered by today's lesson (misconception still present — address at the start of Lesson 3); (b) questions that anticipate

Measurement Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Using the General Absolute Value Equation and the Graphing Calculator (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	the class DQB under the column 'Questions we now have'. The teacher reads three or four questions aloud and briefly sorts them into emerging themes visible on the board: 'Questions about careers', 'Questions about how scientists prove things', 'Questions about the environment'. Learners copy the class driving question into their exercise books and write beneath it: 'Today I learned… and now I wonder…'	write ONE question that is now burning in your mind — something today's research made you curious about.' Allow 2 min. 2. Collect and post notes on the DQB. Read four aloud (choose ones that foreshadow Lessons 3–8 — e.g., 'How does a scientist prove something they cannot see?', 'What careers use General Science in Kenya?', 'How do we know if evidence is reliable?'). 3. Say: 'Look at these questions — do you notice that many of them are really asking: how does a scientist know something is true? That is exactly the direction we are heading.' Point to the class driving question on the board and read it together. 4. Say: 'In your exercise book, under your annotated prediction, write: Today I learned… and now I wonder… — two sentences only, one minute.' 5. Cold-call two learners to share their 'I wonder' statement. Affirm connections to the DQB.	the class's collective curiosity visible and creates accountability — learners can return to the board in later lessons to see which questions have been answered. The 'Today I learned… now I wonder…' sentence frame reinforces metacognitive awareness of the boundary between current knowledge and the unknown.	Lessons 3–5 on inference (good — acknowledge in Lesson 3 opener); (c) questions outside the sub-strand scope (park on a 'Parking Lot' section of the DQB and note which learners wrote them — potential enrichment candidates). Count how many learners wrote a genuine question vs. a statement — those writing statements need prompting on question formation.
Model Building Phase 📹 **VIDEO:** Scale and Indirect Measurement Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Using the General Absolute Value Equation and the Graphing Calculator (Flexbook) Source: CK-12 🔍 Search ARES for similar	Learners open to a fresh page in their exercise books titled 'My Growing Model of General Science'. They draw a simple mind-map with 'General Science' at the centre. From today's lesson they add at least four branches: (1) a definition in their own words, (2) three community connections (food/health/environment/technology), (3) the Kisumu farmer example labelled 'indirect evidence', and (4) one question they still have. This is an initial model — explicitly incomplete. Learners label it 'Model Version 1 — Lesson 2' so they can track how it grows across the eight lessons.	1. Say: 'Scientists do not just collect facts — they build models: pictures of how they think the world works. You are going to start your own model of General Science today. Open a fresh page. Title it: My Growing Model of General Science — Version 1.' 2. Demonstrate on the board: draw a circle in the centre labelled 'General Science', then add two branches as examples ('Definition' and 'Kisumu Farmer — indirect evidence'). Say: 'You will add at least two more branches from your own research today. This model is NOT finished — it will grow every lesson.' 3. Allow 4 min for learners to construct their mind-maps. Circulate and comment	Cumulative Mind-Map Modelling (versioned): the explicitly versioned mind-map externalises the learner's current mental model and makes revision tangible. By labelling it 'Version 1', the teacher signals that science is an iterative, self-correcting process — a theme central to the driving question. Connecting the Kisumu farmer to the definition branch from the first lesson ensures the anchoring phenomenon remains the conceptual thread of every model revision.	At the end of the lesson, do a rapid 'gallery walk' of three or four open exercise books (with permission). Check for: (a) a working definition that goes beyond 'General Science is a subject'; (b) at least one Kenyan-specific community connection (e.g., 'tea farming in Kericho needs soil chemistry'); (c) the Kisumu farmer explicitly present; (d) at least one genuine question on the model. Learners whose models lack the farmer connection need a prompt at the start of Lesson 3: 'Before we begin — how does what you drew yesterday connect to the woman in Kisumu?'

readings		specifically: 'Good — can you add an arrow from the farmer branch to your definition branch to show they are connected?' / 'I see you put health as a branch — can you write one Kenyan example on that branch?' 4. Close the lesson: 'Your model today has four branches. By Lesson 8 it will have many more. Each lesson, we will come back and add to it. Keep this page clean — it is your scientific model, not just notes.'		

D. TEACHER REFLECTION
After the lesson, consider the following: Did most learners move beyond a textbook definition to articulate a personally meaningful, locally grounded understanding of General Science? Were the community connections genuinely Kenyan (specific crops, places, careers) or generic? Did the Kisumu farmer appear organically in group presentations, or did learners treat her as a separate story? If the farmer was not referenced by any group during the Explain phase, plan a stronger bridging prompt for the opening of Lesson 3. Review the DQB sticky notes before Lesson 3: count how many questions are about proof, evidence, or knowing — these signal readiness for the inference principle. If fewer than half the class posted questions about how scientists know things, build more explicit bridging in Lesson 3's Predict phase. Note which learners struggled to produce any community connection during research — these learners may lack background experience with science contexts and will benefit from being paired with a resource-rich partner in Lesson 3's observation activity.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners researched the meaning of General Science using digital devices and/or print media, and presented at least three ways General Science connects to community life in Kenya (food and agriculture, health, environment, technology).
What did I learn?	General Science is not merely a collection of facts — it is a systematic way of asking questions and finding answers using evidence. The Kisumu maize farmer, who inferred a hidden soil pest from the pattern of wilting without seeing the pest directly, is practising General Science.
How does this explain the phenomenon?	Learners co-constructed a class definition of General Science, annotated their opening predictions to show conceptual change, posted new questions on the Driving Question Board, and built Version 1 of a cumulative mind-map model that will grow across all eight lessons in this sub-strand.

LESSON 3 (40 minutes): Importance of General Science in Life, Environment and Technology	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners explore the importance of General Science in human life, the environment, and technology by researching real Kenyan examples — and connect each example back to the core idea that science allows us to infer what we cannot directly see.
Knowledge	Learners outline the importance of General Science in life (nutrition, disease), environment (conservation, pollution monitoring), and technology (mobile money, water treatment, food safety).
Skills	Learners use available digital and/or print media to research on the importance of General Science in relation to human life, environment and technology; discuss with peers; and present findings in plenary.
Attitudes	Learners appreciate the importance of General Science in day-to-day life and develop respect for others' opinions when discussing the importance of General Science.
Key Inquiry Question	How is General Science useful in daily life?
Purpose in Storyline	In Lesson 1, learners encountered the anchoring phenomenon — a Kisumu maize farmer who correctly inferred a soil pest she never saw. In Lesson 2, learners explored the meaning of General Science and the principle of inference. Now in Lesson 3, learners widen their lens: they gather evidence — through peer discussion and media research — that General Science operates everywhere in Kenyan life. Every example they find (a blood test, NEMA air-quality sensors, M-PESA's engineering backbone) will reinforce the same insight: scientists and technologists routinely make reliable conclusions about things they cannot directly observe. This advances the Driving Question Board by adding new real-world domains to the class's growing answer.
Safety Notes	No hazardous materials are used in this lesson. If learners use personal digital devices, remind them to observe responsible and safe internet use (cyber security). Learners must not share personal information online. Discussions should remain respectful; the teacher monitors for any exclusionary language during group work.

B. LESSON OVERVIEW

Learners discuss with peers the importance of General Science in three domains — human life, environment, and technology — then use print or digital media to find one concrete Kenyan example per domain. Groups present their examples in a brief gallery-style plenary. The teacher anchors every example to the phenomenon: in each case, someone had to infer something they could not directly see. The lesson closes with an update to the Driving Question Board and a cumulative model revision.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [When to use z or t statistics in]	Individually, learners write a silent prediction in their science notebooks: 'List three ways you think General Science affects your	Teacher writes the three-category frame on the board: HUMAN LIFE | ENVIRONMENT | TECHNOLOGY. Teacher says:	Silent Prediction + Star-Marking: Learners activate prior knowledge and self-assess which examples already connect to the inference	Teacher scans notebooks during circulation. Observable indicator: learner has written at least one example per category AND has

significance tests Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Using the General Absolute Value Equation and the Graphing Calculator (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	daily life in Kenya — one connected to your body or health, one connected to the environment around you, and one connected to a technology you use.' Learners are then asked: 'Which of your three examples do you think involves a scientist figuring out something they could NOT directly see? Put a star next to it.' After 2 minutes of silent writing, learners share their starred example with a shoulder partner for 1 minute.	'Before we look at any evidence today, I want to know what YOU already think. You have two minutes — silent, independent writing. Go.' [Allow 2 minutes silent think time.] Teacher circulates, scanning notebooks without commenting. After partner share, teacher cold-calls 3 learners (one per category) using name cards: 'Akinyi, what did you star for human life — and why did you think it involves something invisible?' [WAIT TIME: 12 seconds after each cold-call before prompting.] Teacher does NOT confirm or correct predictions — instead says: 'Hold that thought. By the end of today's lesson you will be able to test your own prediction.'	theme. This creates a personal stake in the evidence-gathering phase that follows.	placed a star on at least one. Teacher notes any learner who cannot generate a technology example — this signals a gap to address during the Observe Phase.
Observe Phase 📹 **VIDEO:** When to use z or t statistics in significance tests Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Using the General Absolute Value Equation and the Graphing Calculator (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners work in groups of four. Each group is assigned one of six Kenyan 'science-in-action' scenario cards (prepared as printed strips or displayed on the class screen): (A) KEBS tests maize flour for aflatoxin — the poison is invisible to the naked eye; (B) A doctor at Kenyatta National Hospital reads a chest X-ray to detect TB — the bacteria are not directly visible; (C) NEMA deploys air-quality sensors along Mombasa Road, Nairobi, to detect invisible pollutants; (D) Kenya Wildlife Service rangers count elephants in Tsavo using camera traps and footprint analysis — most elephants are never directly seen; (E) Engineers at Safaricom built M-PESA on radio-wave technology — the signals that carry money are invisible; (F) Nairobi Water and Sewerage Company tests treated water for invisible bacteria before it reaches	Teacher distributes or displays the six scenario cards. Teacher reads Card A aloud as a model, then thinks aloud: 'KEBS scientists cannot see aflatoxin in ugali flour with their eyes — they infer it is present from a chemical colour-change test. That is the same kind of thinking our Kisumu farmer used.' Teacher then releases groups to work: 'You have 12 minutes. First classify your card, then find your own Kenyan example. Use the two-column table.' [Sets visible timer for 12 minutes.] Teacher circulates, stopping at each group for 60–90 seconds. Probing questions: 'What exactly cannot be seen?' / 'How does the scientist know it is there if they cannot see it?' / 'Is this human life, environment, or technology — or could it be more than one?' If a group finishes early, teacher challenges: 'Can you find a Kenyan scientist by name who	Scenario Card + Two-Column Evidence Table: The scenario cards provide concrete Kenyan anchors so that the abstract concept of 'importance of General Science' becomes specific and local. The two-column structure (WHAT SCIENCE DOES | WHAT CANNOT BE SEEN DIRECTLY) directly embeds the inference principle from Lesson 2 into the new content, preventing learners from treating this as a disconnected list-making exercise.	Observable indicator during circulation: the second column of each group's table contains a specific invisible entity (e.g., 'aflatoxin molecules', 'TB bacteria', 'carbon monoxide gas', 'elephant population size', 'radio waves', 'E. coli bacteria') — NOT a vague answer like 'science things'. If the second column is blank or vague, teacher asks: 'What exactly is the thing they cannot see with their eyes alone?' and waits 10 seconds.

	homes. Each group reads their card, discusses: 'What domain is this — human life, environment, or technology?' and 'What is the thing the scientist or engineer cannot directly see, but must infer?' Groups then use available print media (newspapers, textbooks) or digital devices to find one additional local Kenyan example in their domain. Groups record findings in a two-column table: WHAT SCIENCE DOES | WHAT CANNOT BE SEEN DIRECTLY.	works in this domain?' [WAIT TIME: 10 seconds minimum after each probing question before offering a hint.]		
Explain Phase 📹 **VIDEO:** When to use z or t statistics in significance tests Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Using the General Absolute Value Equation and the Graphing Calculator (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each group presents their scenario card example plus their self-found Kenyan example in a 90-second gallery share. The class uses a listening protocol: one learner from each audience group must be ready to answer, 'What could NOT be seen directly in that example?' After all six presentations, the full class discusses: 'What pattern do you notice across all six examples — human life, environment, AND technology?' Learners individually write one sentence in their notebooks completing the stem: 'General Science is important because in every domain — life, environment, and technology — scientists must infer _________ that they cannot directly observe, such as _________.'	Teacher facilitates the gallery share using a visible 90-second timer per group. After each presentation teacher asks the audience a cold-call question — rotating through different learners each time: 'Kamau, in this example from Group C — what was the thing NEMA could not see directly?' [WAIT TIME: 12 seconds.] After all six presentations, teacher writes on the board: 'PATTERN?' and asks: 'Turn and talk for 45 seconds — what single idea connects ALL six examples?' After the turn-and-talk, teacher cold-calls two learners and then anchors the pattern explicitly: 'In every single case — whether it is a doctor in Nairobi, a NEMA engineer on Mombasa Road, or a Safaricom technologist — science gives us the power to know something is true even when we CANNOT see it directly. That is exactly what our Kisumu maize farmer did. She was doing science.' Teacher writes on the board: GENERAL SCIENCE = RELIABLE KNOWLEDGE FROM INVISIBLE EVIDENCE. Teacher then connects to KICD outcomes verbatim: 'This is why the	Pattern Recognition across Domains + Sentence Stem Consolidation: By hearing six distinct Kenyan examples and then being asked to identify a single unifying pattern, learners perform the cognitive work of abstraction — moving from specific cases to a general principle. The sentence stem forces each learner to articulate the inference principle in their own words, consolidating the cross-lesson thread.	Teacher collects or scans the completed sentence stems. Observable indicator of meeting expectations: the sentence names a specific invisible entity AND connects it to one of the three domains. Observable indicator of exceeding expectations: the sentence references the inference principle explicitly (e.g., 'scientists infer from indirect evidence'). Learners who write only 'scientists study things' have not yet connected to the phenomenon — teacher notes these for targeted questioning in Lesson 4.

		curriculum tells us General Science helps us explain the importance of General Science in life, environment, and technology.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** When to use z or t statistics in significance tests Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Using the General Absolute Value Equation and the Graphing Calculator (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each group writes one sticky note (or one entry on the class DQB section of the board) in response to: 'What new evidence do we now have that helps answer the Driving Question: How does a scientist know something is true when they cannot see it directly?' Groups also add one new question their research raised — e.g., 'If NEMA uses sensors, who decides what level of pollution is dangerous?' or 'Can science ever be wrong about something invisible?' Learners read each other's new questions and place a tick next to any question they also have.	Teacher directs groups to the DQB wall/section. Teacher reminds learners of the DQB protocol established in Lesson 1: 'Green sticky = new evidence that moves us toward an answer. Orange sticky = new question we now have.' Teacher models one entry: 'Here is mine — green: Scientists at KEBS use chemical reactions to infer aflatoxin in maize flour — they never see the toxin directly, but they know it is there. That is evidence for our driving question.' Teacher gives groups 3 minutes to write and post. After posting, teacher reads two or three new questions aloud and says: 'These are brilliant scientific questions. We will come back to them as we move through the sub-strand.' Teacher updates the class's running 'Evidence We Have' column on the DQB by adding: 'General Science allows inference in three domains: human life, environment, technology — all confirmed by Kenyan examples.'	Driving Question Board — Evidence + New Questions Protocol: The DQB makes the class's growing understanding visible and cumulative. Distinguishing 'evidence' from 'new questions' teaches learners that science is an iterative process — answering one question opens new ones. The tick-voting on questions builds community ownership of the inquiry.	Observable indicator: each group's green sticky contains a specific Kenyan science example AND explicitly states what was inferred (not directly seen). If a group's sticky only says 'science is useful in Kenya' without specifying inference, teacher prompts: 'Can you add — what exactly could not be seen directly in your example?'
Model Building Phase 📹 **VIDEO:** When to use z or t statistics in significance tests Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Using the	Learners retrieve the personal inference model they began building in Lesson 2 (a simple diagram or concept map showing: HIDDEN CAUSE → INDIRECT EVIDENCE → INFERENCE → CONCLUSION). They now expand the model by adding at least two new real-world 'arms' — one from each domain they explored today (e.g., an arm labelled ENVIRONMENT showing: invisible pollutant → sensor reading → inference → NEMA action; and an	Teacher says: 'Open your model from Lesson 2. Today you discovered that our farmer's way of thinking is not unique — it happens in hospitals, conservation parks, and on your phone. Expand your model — add two new arms, one for environment and one for technology, using your Kenyan examples from today. You have 5 minutes.' [Sets timer.] Teacher circulates and looks for: (1) is the model genuinely expanded, not just re-labelled? (2) does each	Cumulative Model Revision (Progressive Elaboration): Returning to and expanding the same model across lessons makes learning visible and non-fragmented. Each new arm learners add is a concrete record of growing understanding. The structural constraint (HIDDEN CAUSE → INDIRECT EVIDENCE → INFERENCE → CONCLUSION) ensures that the inference principle remains the organising spine of the model, not	Observable indicator of meeting expectations: model has at least two new arms, each with a specific Kenyan context and the four-step inference chain. Observable indicator of approaching expectations: new arms added but the 'hidden cause' box is missing or vague. Teacher notes these learners for a brief check-in at the start of Lesson 4.

General Absolute Value Equation and the Graphing Calculator (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	arm labelled TECHNOLOGY showing: invisible radio wave → M-PESA transaction → inference → money transferred). Learners annotate each new arm with the Kenyan context. If time permits, one volunteer shares their expanded model with the class under the visualiser or by holding it up.	new arm follow the HIDDEN CAUSE → INDIRECT EVIDENCE → INFERENCE → CONCLUSION structure? Teacher prompts where needed: 'What is the hidden thing in your technology example? Where does it go on the model?' If a volunteer shares, teacher asks the class: 'Does their model correctly show that the evidence is INDIRECT? Give a thumbs up if you agree.' Teacher closes by saying: 'Next lesson, Lesson 4, we will look at careers built on exactly this skill. You will need your model — keep it safe.'	mere content accumulation.	

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did learners' sentence stems in the Explain Phase show genuine understanding of the inference principle, or did they default to listing science facts? If most stems were vague, plan a brief re-anchor at the start of Lesson 4 using one of the Kenyan scenario cards as a quick warm-up. (2) Which of the six scenario cards generated the richest discussion — and which fell flat? Consider whether the card was too abstract or whether the Kenyan context was not close enough to learners' lived experience. For the next cohort, replace weaker cards with examples closer to learners' home counties. (3) Were all three domains — human life, environment, technology — represented in the DQB green stickies? If one domain was under-represented, assign it as a specific focus for the independent media research task at the start of Lesson 4. (4) Did any learner's expanded model in Phase 5 show a genuinely creative new arm beyond the two required? Note these learners — they may be ready for the extension challenge of drawing a single unified model that connects all three domains. (5) Were digital devices used responsibly? If any cyber safety concerns arose, address them explicitly in the next lesson's opening.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners examined six Kenya-context scenario cards showing General Science at work in human health, environmental monitoring, and technology — including KEBS aflatoxin testing in maize flour, NEMA air-quality sensing on Mombasa Road, KWS wildlife inference in Tsavo, KNH chest X-ray reading, Safaricom M-PESA radio-wave engineering, and Nairobi Water bacterial testing. In every case, the key action was the same as the Kisumu farmer's: inferring a hidden, invisible entity from indirect evidence.
What did I learn?	General Science operates across three major domains of Kenyan life — human life (nutrition, disease), environment (conservation, pollution), and technology (communications, water treatment, food safety). In all three domains, the defining power of science is the ability to make reliable conclusions about things that cannot be directly observed, using indirect evidence and inference. This is not exceptional — it is the everyday work of scientists, doctors, engineers, and environmental officers across Kenya.
How does this explain the phenomenon?	The reason General Science is important is not simply that it produces useful facts — it is that it provides a systematic, evidence-based method for knowing what is real even when direct observation is impossible. The Kisumu farmer, the KEBS food-safety technologist, the NEMA engineer, and the Safaricom radio engineer are all practising the same core scientific habit: collecting indirect evidence, recognising patterns, and inferring a hidden cause. This principle — inference from indirect evidence — is the

	thread that connects the meaning of General Science (Lesson 2) to its importance in life, environment and technology (Lesson 3), and will next connect to the careers built on this skill (Lesson 4).

LESSON 4 (80 minutes (2 × 40-minute lessons)): Careers in General Science: Research, Chart Making & Career-Path Justification	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners identify career opportunities connected to General Science sub-topics, construct personal career-path charts, and justify their chosen path — building the understanding that General Science is not merely abstract knowledge but a direct gateway to real Kenyan livelihoods and professions.
Knowledge	Learners will know: (a) the broad sub-topics of General Science (biology, chemistry, physics) and the specific career pathways each unlocks; (b) the names and roles of key science-based careers in Kenya (doctor, clinical officer, agronomist, lab technologist, environmental scientist, teacher, engineer, pharmacist, marine & fisheries technologist, food scientist); (c) how indirect evidence and inference — the skill anchored in the Kisumu maize-farmer phenomenon — are actively used in each of these careers daily.
Skills	Learners will be able to: (a) use digital devices or print media to research career opportunities related to General Science; (b) draw and annotate a personal career-path chart linking General Science sub-topics to specific Kenyan careers; (c) write a concise 3-sentence career justification using evidence-based reasoning; (d) communicate career choices clearly in a cold-call share-out.
Attitudes	Learners will appreciate: (a) the direct relevance of General Science to their own futures and to Kenya's socio-economic development; (b) the value of self-efficacy — believing they can pursue a science career regardless of background; (c) responsibility in using digital devices appropriately during research; (d) respect for peers' diverse career choices.
Key Inquiry Question	How is General Science useful in daily life?
Purpose in Storyline	In Lessons 1–3 learners encountered the Kisumu maize-farmer phenomenon, explored what General Science is, examined its importance, and began mapping scientific processes. Lesson 4 is the first deep Observe Phase focused entirely on careers: learners now gather real evidence — through structured research — of who uses General Science professionally in Kenya and how. This advances the driving question by showing that the farmer's inferential reasoning is not unique to farming; it is the heartbeat of every science career. The career-path charts produced here will be referenced again in the Model Building phase (Lessons 7–8) when learners revise their master model of 'What General Science is and how scientists know.'
Safety Notes	No hazardous materials are used. When using digital devices, learners must observe cyber-safety rules: use only school-approved websites (e.g., Kenya Medical Training College site, KEPHIS, NEMA, universities' career pages, Kenya Education Cloud). Learners must not share personal information online. Chart-making materials (scissors, rulers, markers) must be handled with care. If a resource person visits, learners must listen respectfully and not record audio/video without permission.

B. LESSON OVERVIEW

Learners open by re-visiting the Kisumu maize-farmer scenario and predicting which science careers use the same kind of inferential reasoning she used. They then enter a structured Observe Phase: using digital devices or print media (career booklets, newspaper cuttings, KICD resources) to research at least four science-linked Kenyan careers. Each learner produces a hand-drawn or digital career-path chart that maps three General Science sub-topics (biology, chemistry, physics) onto specific careers, annotating each career with the type of evidence or inference it relies upon — thus looping back to the phenomenon. Learners select one career, write a 3-sentence justification, and display charts on the classroom wall ('Career Gallery'). The teacher conducts a cold-call share-out of five learners (8 seconds wait time each). The lesson closes with a Driving Question Board (DQB) update and a 2-minute silent reflection exit slip.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Prime numbers] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Using the General Absolute Value Equation and the Graphing Calculator (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners re-read the Kisumu maize-farmer scenario displayed on the board (or read aloud by a volunteer). They then individually write on a sticky note or in their science journals: 'Name TWO careers where a person must draw conclusions from evidence they cannot directly see — just like the farmer did.' Learners share with a shoulder partner for 2 minutes, then the teacher takes four responses from the class. Learners also complete a quick 3-column KWL strip (Know / Want to Know / Learned) for 'Careers in General Science' — only the K and W columns are filled at this stage.	1. Display the maize-farmer scenario on the board or read it aloud: 'A maize farmer in Kisumu notices her plants along one edge of the shamba are wilting and yellowing — she has never seen any insect, yet she correctly concludes a soil pest is destroying the roots. How did she know?' Ask: 'Before we research today, I want you to think — who else in Kenya makes conclusions from things they cannot directly see? Write two careers on your sticky note. You have 90 seconds. Go.' [WAIT: allow 90 seconds of silent individual writing.] 2. Say: 'Turn to your shoulder partner. Share your two careers and explain WHY that career needs inference. You have 2 minutes.' [WAIT: 2 minutes pair talk.] 3. Cold-call FOUR learners using a random name selector or class list — do NOT accept hand-raises for this opener. For each learner say: 'Tell us your career and one sentence about the inference they make.' [WAIT TIME: 8 seconds after naming each learner before prompting.] 4. Anchor responses on the board under the heading 'Careers that use inference — like our farmer.' Leave this anchor visible all lesson. 5. Distribute or project the KWL strip. Say: 'Fill in the K and W columns only. What do you already KNOW about science careers in Kenya? What do you WANT to know? Leave the L column blank — you will fill it at the end.' [WAIT: 2 minutes.]	KWL Strip + Sticky-Note Prediction: This activates prior knowledge and creates cognitive dissonance — most learners know science is 'useful' but have not consciously mapped it to careers. By anchoring predictions to the phenomenon (farmer's inference), the strategy ensures career exploration is not disconnected but is framed as 'who else does what the farmer did?'	Observable indicator: Each learner has at least two careers written on their sticky note before pair-share begins. Teacher scans the room during the 90-second writing window. During cold-call, listen for whether learners can articulate an inferential action (e.g., 'a doctor reads an X-ray to find a broken bone they cannot touch' — not just naming the career). Flag any learner who cannot articulate inference for targeted support in Phase 2.
Observe	Learners work individually (or in	1. Before research begins, say:	Structured Evidence Table (Career	Observable indicator: After 18

Phase 📹 **VIDEO:** [Prime numbers] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Using the General Absolute Value Equation and the Graphing Calculator (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	pairs if devices are limited) to research career opportunities related to General Science using available digital devices (smartphones, tablets, school computers, Kenya Education Cloud) or print media (career booklets, newspaper cuttings, textbook appendices, KICD resource sheets). Each learner must find and record information on AT LEAST FOUR careers across the three General Science sub-topic areas: biology-linked (e.g., doctor, nurse, clinical officer, agronomist, food scientist, marine & fisheries technologist), chemistry-linked (e.g., pharmacist, lab technologist, chemical engineer, environmental scientist), and physics-linked (e.g., civil engineer, radiographer, meteorologist, electronics technician). For each career, learners record: (a) career name, (b) science sub-topic(s) it draws from, (c) one Kenyan institution or county where this career is practised (e.g., Kenyatta National Hospital – Nairobi, KEPHIS – Nairobi, NEMA field offices – Rift Valley, KMFRI – Kisumu/Mombasa), and (d) ONE example of how that professional uses inference or indirect evidence. Research time: 18 minutes.	'You are now detectives — just like our Kisumu farmer. You are going to gather evidence about science careers in Kenya. You must find at least four careers, and for EACH one you must answer this question: How does this professional draw a conclusion from something they cannot directly see? Write your findings in your science journal under the heading: Career Evidence Table.' Display the four-column table heading on the board: Career Name | Science Sub-topic | Kenyan Institution/County | How they use inference. 2. Distribute devices or print resources. For print-only classrooms, pre-prepare a one-page Career Resource Sheet with brief profiles of 8 Kenyan science careers (doctor at KNH, agronomist at Mwea Irrigation Scheme, lab technologist at KEMRI, environmental scientist at NEMA, marine technologist at KMFRI Kisumu, pharmacist at Aga Khan Hospital Nairobi, meteorologist at Kenya Meteorological Department, physics teacher at a county secondary school). 3. Circulate every 3 minutes. Stop at six learners individually. Ask quietly: 'What is one career you have found? How does that person use inference?' [WAIT: 8 seconds each.] If a learner is stuck, prompt: 'Think about what the farmer did — she used patterns and absence of other causes. Does your career professional do something like that?' 4. At the 15-minute mark, give a 3-minute warning: 'You have 3 minutes left. Make sure you have filled in all four columns for at least four careers.' 5. Cyber-safety reminder at the start: 'Use only	Evidence Table): By requiring learners to explicitly state HOW each career uses inference, the research activity is not passive information collection — it is evidence-to-phenomenon mapping. This mirrors the NGSS practice of 'Obtaining, Evaluating, and Communicating Information.' The four-column structure scaffolds learners who struggle to move beyond surface career facts toward deeper sensemaking about scientific reasoning in professional life.	minutes, each learner's science journal has at least four rows in the Career Evidence Table, with the 'How they use inference' column containing a specific action verb (e.g., 'analyses,' 'interprets,' 'measures,' 'detects') — not just a vague phrase like 'uses science.' Teacher checks this during circulation and flags any journal with fewer than three rows or empty inference column for immediate scaffolding.

		school-approved websites. Do not share personal information. If you find a site you are unsure about, raise your hand.'		
Explain Phase 📹 **VIDEO:** [Prime numbers](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Using the General Absolute Value Equation and the Graphing Calculator (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Using their Career Evidence Table, each learner draws a personal career-path chart on A4 paper (or digitally). The chart must show: (a) three branches labelled Biology, Chemistry, Physics at the top; (b) at least two careers branching from each sub-topic; (c) arrows showing how General Science knowledge flows into the career; (d) a star (★) next to the ONE career they identify with most. Below the chart, learners write exactly 3 sentences justifying their chosen career, using this sentence frame as a guide — Sentence 1: 'I am interested in [career] because…'; Sentence 2: 'This career connects to General Science through [sub-topic] because…'; Sentence 3: 'Like the Kisumu maize farmer, a [career name] uses indirect evidence when they…' Charts are displayed on the classroom wall as a 'Career Gallery.' Learners do a 5-minute silent gallery walk, placing a small tick on two peers' charts whose inference example they find most convincing.	1. Say: 'You now have the evidence. It is time to make sense of it by building your career-path chart. This chart is YOUR map — it shows how General Science connects to your future. You have 15 minutes to draw and complete it. The most important part is the third sentence of your justification — it must connect your career to what the Kisumu farmer did. If you cannot write that sentence, your chart is incomplete.' 2. Display on the board: the 3-sentence frame (see learner experience above). Read it aloud once. 3. Circulate. Stop at learners who are writing the third sentence. Ask: 'How does your career professional make a conclusion without directly seeing something? Give me a specific example from Kenya.' [WAIT: 8 seconds.] Examples to model if needed: 'A lab technologist at KEMRI tests a blood sample and infers which disease a patient has — they never see the bacteria with the naked eye.' / 'An agronomist at the Rift Valley Agricultural Research Station tests soil pH and infers which nutrients are deficient — they never see the nutrients directly.' 4. At 15 minutes, instruct learners to display charts on the wall. Say: 'Stand up. Take your chart to the wall. Put your name at the bottom. Then do a silent gallery walk for 5 minutes — place a small tick on TWO charts whose inference example you find most convincing or most creative.' [WAIT: allow 5-minute silent gallery walk.] 5. After the gallery	Career-Path Chart + 3-Sentence Phenomenon-Linked Justification + Peer Gallery Walk: The chart externalises internal reasoning, making scientific career structures visible and comparable. The mandatory third sentence forces a direct cognitive link back to the anchoring phenomenon — preventing careers from becoming an isolated list-making exercise. The gallery walk activates peer assessment and social learning, and the tick system gives every learner low-stakes feedback on the quality of their inference example. This combination enacts the NGSS practice of 'Constructing Explanations' at a personal, career-relevant level.	Observable indicators during cold-call share-out: (1) The learner names a specific Kenyan institution or location connected to their career (e.g., 'I want to be an environmental scientist working at NEMA in Nairobi' — not just 'environmental scientist'). (2) The third sentence contains a specific indirect evidence action (e.g., 'measures water turbidity in Lake Victoria to infer pollution levels' — not 'uses science to help'). Teacher records on a simple 3-column checklist: Learner name | Career named with Kenyan context (Y/N) | Inference action specific (Y/N). Any learner scoring N on both is flagged for the exit slip follow-up.

		walk, cold-call FIVE learners by name (use class list, row 1 and row 3 alternating) to share their chosen career and read their three sentences aloud. For each, say: '[Name], please stand and share your career and your three sentences.' [WAIT TIME: 8 seconds after naming, before the learner begins.] After each share, ask the class: 'Does this career use inference like the farmer? Thumbs up if yes, thumbs sideways if you are not sure.' Address any thumbs-sideways responses with a clarifying question to the class.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Prime numbers Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Using the General Absolute Value Equation and the Graphing Calculator (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class returns to the Driving Question Board displayed at the front of the room. The teacher reads the original driving question aloud: 'How does a scientist know something is true when they cannot see it directly — and how is this power of General Science shaping life, careers, and the environment in Kenya?' Learners are given two sticky notes each. On the FIRST (green) sticky note, they write one new thing they now understand about this question based on today's lesson — specifically about careers. On the SECOND (orange) sticky note, they write a new question that today's research has raised for them — a question they still cannot answer. Both sticky notes are placed on the DQB. The teacher reads three orange sticky-note questions aloud and posts them under the new DQB sub-heading: 'What science skills do I need for my chosen career?' This sub-heading remains on the DQB for Lessons 5–8.	1. Point to the DQB and read the driving question aloud slowly. Say: 'Every lesson we update this board. Today we are going to add what we have learned AND what we are still wondering. You have two sticky notes.' [Distribute sticky notes.] 2. Say: 'On the GREEN note — write one sentence beginning with: I now understand that… Connect it to careers and to the farmer if you can. On the ORANGE note — write one question beginning with: I am still wondering… This should be a question today's research raised but did not answer. You have 3 minutes.' [WAIT: 3 minutes silent writing.] 3. Invite learners to post both notes on the DQB (green on the left, orange on the right). 4. Read THREE orange notes aloud — choose ones that reflect genuine scientific thinking (e.g., 'I am still wondering how an agronomist decides which soil test to use when there are many possible problems' or 'I am still wondering if there are science careers specific to Lake Victoria	Two-Colour Sticky-Note DQB Protocol (Green = Understanding Built / Orange = New Questions): This strategy makes metacognition visible and communal. The green notes consolidate today's evidence; the orange notes fuel the inquiry forward — modelling how real scientists always generate new questions from new evidence. By publicly naming the new DQB sub-question ('What science skills do I need for my chosen career?'), the teacher ensures learners carry a personal stake into the remaining lessons. This enacts the NGSS practice of 'Asking Questions and Defining Problems.'	Observable indicator: Each learner's green sticky note contains the word 'career' AND at least one of the following: a career name, a Kenyan location, the word 'inference' or 'evidence,' or a reference to the farmer. Any green note that is blank or contains only a career name with no connection to scientific reasoning is flagged. Teacher collects and photographs all DQB notes at end of lesson for review.

		fishing communities'). After reading each, say: 'This is an excellent scientific question. We will come back to it.' 5. Write the new DQB sub-heading in large text: 'What science skills do I need for my chosen career?' and post it prominently. Say: 'This question will guide us for the rest of the unit. Keep your career chart — you will need it in Lessons 7 and 8.'		
Model Building Phase 📹 **VIDEO:** Prime numbers Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Using the General Absolute Value Equation and the Graphing Calculator (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their cumulative science model — the growing diagram of 'What General Science is and how scientists know' begun in Lessons 1–3. They add a NEW LAYER to their model today: a 'Careers Branch.' Using a different-coloured pen, each learner draws three to five career icons (simple stick-figure sketches or written labels) around the outside of their model and draws arrows from the relevant General Science sub-topics (biology, chemistry, physics) to each career. On each arrow, learners write a ONE-WORD inference action (e.g., 'analyses,' 'measures,' 'tests,' 'detects,' 'infers'). Finally, learners draw one arrow from their chosen career back to the centre of the model — labelled 'uses indirect evidence like the farmer' — to reinforce the phenomenon connection. In the last 2 minutes, learners complete the L column of their KWL strip from Phase 1.	1. Say: 'Take out your cumulative science model. Today we add the Careers Branch. Use a NEW colour so we can see what grew today. You have 7 minutes.' [Distribute coloured pens if available.] 2. Circulate. For each learner you stop at, ask: 'Show me the arrow going back to the centre of your model. What does it say?' [WAIT: 8 seconds.] If the arrow is missing, prompt: 'Remember our farmer? Every career you found today uses the same core skill she used. Can you show that connection on your model?' 3. At 5 minutes, say: 'One minute left on the model. Make sure you have at least one arrow going BACK to the centre saying indirect evidence or inference.' 4. After model building, say: 'Now return to your KWL strip. Fill in the L column: what did you LEARN today about careers and General Science? You have 2 minutes.' [WAIT: 2 minutes.] 5. Close the lesson: 'Today you gathered real evidence — just like a scientist. You found that careers from Kisumu to Mombasa to Nairobi are all built on the same skill our farmer used: drawing reliable conclusions from evidence you cannot directly see. In the next lesson, we will go deeper — we will look at how scientists formally	Cumulative Model Revision with Careers Branch (Different-Colour Protocol): The colour-differentiation protocol makes the growth of understanding literally visible — learners and teacher can see exactly what conceptual layer was added in Lesson 4 versus earlier lessons. Adding the reverse arrow back to the phenomenon centre prevents the Careers Branch from becoming a disconnected list; it is anchored to the sub-strand's core idea. Completing the KWL L-column closes the metacognitive loop opened in Phase 1. This enacts the NGSS practice of 'Developing and Using Models.'	Observable indicators: (1) The cumulative model has a visibly distinct Careers Branch in a new colour with at least three careers labelled. (2) At least one arrow on the model points back to the centre with an inference/indirect evidence label. (3) The exit slip (collected at the door) contains a specific evidence type AND what it reveals — not a generic statement. Teacher sorts exit slips into three piles overnight: Secure (specific evidence + reveals), Developing (career named but inference vague), Beginning (no inference connection). Developing and Beginning learners receive a targeted prompt card at the start of Lesson 5.

		use inference and indirect evidence as a scientific principle. Keep your model. Keep your chart. Both are growing.' 6. Exit Slip (written, collected at door): 'In one sentence, explain how YOUR chosen career uses indirect evidence. Be specific — name the evidence and what it reveals.' Collect as learners leave.		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal:

1. PHENOMENON CONNECTION: During the cold-call share-out of five learners, how many of the five third sentences made a specific, accurate connection to indirect evidence or inference — not just a general reference to 'using science'? If fewer than three did, what scaffold could you add to the sentence frame in a future lesson?

2. EQUITY & ACCESS: Were learners without digital device access able to gather equally rich career information from the print Career Resource Sheet? Did the sheet need more Kenyan-specific career profiles (e.g., careers specific to the Coast, Nyanza fishing communities, or the Rift Valley agricultural belt)?

3. CAREER DIVERSITY: Did the career charts show gender balance in career choices? If girls overwhelmingly chose nursing/teaching and boys chose engineering, consider opening Lesson 5 with a brief profile of a Kenyan female scientist (e.g., Prof. Miriam Were — public health, or Dr. Lydia Njenga — molecular biology) to broaden the range of models available to learners.

4. DQB QUALITY: Were the orange sticky-note questions genuinely scientific (i.e., asking about mechanisms, evidence, or processes) or were they social/motivational ('Will I get a good job?')? If mostly motivational, the DQB introduction may need re-framing in Lesson 5 to push toward scientific inquiry questions.

5. MODEL BUILDING: Did the cumulative models show the career-to-centre reverse arrow? If fewer than half of learners drew the reverse arrow, the phenomenon anchor is not yet deeply internalised — plan a whole-class model share at the start of Lesson 5 to reinforce it.

6. EXIT SLIP DATA: Use exit slip piles (Secure / Developing / Beginning) to form flexible groups for the start of Lesson 5. Plan a 5-minute targeted small-group discussion for Beginning learners using the prompt: 'A doctor at Kenyatta National Hospital looks at a blood test result. What are they trying to find out? Can they see it directly? How is this like the Kisumu farmer?'

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners used digital devices or print media (Career Resource Sheets featuring Kenyan science professionals at KNH, KEMRI, NEMA, KMFRI, KEPHIS and county agricultural offices) to research at least four General Science-linked careers in Kenya, recording findings in a structured four-column Career Evidence Table (Career Name / Science Sub-topic / Kenyan Institution or County / How they use inference).
What did I learn?	Learners constructed personal career-path charts mapping biology, chemistry, and physics sub-topics to specific Kenyan careers, wrote a 3-sentence career justification with a mandatory inference-connection sentence linking their chosen career back to the Kisumu maize-farmer phenomenon, and updated the Driving Question Board with a new sub-question: 'What science skills do I need for my chosen career?'
How does this explain the phenomenon?	Through the cold-call share-out (five learners, 8 seconds wait time each) and the cumulative model's new Careers Branch, learners articulated that General Science careers — from agronomist in the Rift Valley to lab technologist at KEMRI to environmental scientist at NEMA — all depend on the same core scientific practice that the Kisumu farmer used: drawing reliable conclusions

	from indirect evidence that cannot be directly seen.

LESSON 5 (40 minutes): Introduction to Scientific Processes and the Principle of Inference (Explain Phase – Part 1)	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To formally introduce the principle of inference as the foundation of scientific reasoning, enabling learners to distinguish evidence and reasoning from direct observation, and to apply the evidence-reasoning-inference framework to real Kenyan contexts.
Knowledge	Learners explain that inference is a conclusion drawn from evidence combined with reasoning, not from direct observation; learners outline the three-part structure of scientific inference (evidence → reasoning → inference); learners identify the principle of inference as foundational to all scientific knowledge.
Skills	Learners apply the evidence-reasoning-inference framework to analyse the Kisumu farmer scenario, a Nairobi doctor scenario, and an Amboseli ranger scenario; learners record a structured three-column table in their exercise books; learners distinguish between what is observed and what is inferred.
Attitudes	Learners appreciate that reliable scientific conclusions can be drawn without direct observation; learners respect the value of systematic reasoning as a tool for understanding hidden phenomena; learners demonstrate responsibility by listening attentively and contributing to peer discussion.
Key Inquiry Question	How is General Science useful in daily life?
Purpose in Storyline	Lessons 1–4 established the anchoring phenomenon (the Kisumu farmer), introduced science as a discipline, explored its importance in daily Kenyan life, and surveyed careers in science. Lesson 5 is the pivotal Explain Phase Part 1: learners now receive the formal conceptual tool — the principle of inference — that explains HOW the farmer knew. This lesson names and structures what learners have been intuitively grappling with, and extends the framework to two new Kenyan phenomena (a Nairobi doctor reading an X-ray; an Amboseli ranger tracking a poacher). By the end, every learner has a written evidence-reasoning-inference table they can use as a cognitive anchor for the rest of the sub-strand.
Safety Notes	No laboratory hazards. Ensure learners do not share personal medical histories when discussing the doctor scenario. Teacher should be sensitive: the Amboseli ranger scenario may surface conservation anxieties — keep discussion focused on scientific reasoning, not political conflict.

B. LESSON OVERVIEW

Learners revisit the Kisumu farmer anchoring phenomenon and formally decode it using the principle of inference — a conclusion drawn from evidence plus reasoning, not from direct observation. The teacher introduces the three-part evidence-reasoning-inference (ERI) framework through direct instruction with interactive questioning, then guides learners to apply the framework to two additional Kenyan phenomena: a Nairobi radiologist reading a chest X-ray, and an Amboseli wildlife ranger identifying a poacher's route from indirect signs. Learners record a structured three-column ERI table for all three scenarios in their exercise books. The lesson closes with a Driving Question Board update and a brief exit card. No materials or laboratory apparatus are required — this is a teacher-led sensemaking lesson anchored entirely in Kenyan real-life contexts.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** When to use z or t statistics in significance tests Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Bathymetric Evidence for Seafloor Spreading (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners read the following prompt displayed or dictated by the teacher: 'Akinyi, the maize farmer in Kisumu, walks along the edge of her shamba early in the morning. She sees her maize plants wilting and yellowing — but she sees NO insect, NO fungus, NO visible damage above the soil. She goes home and tells her daughter: I am sure there is a soil pest destroying the roots.' Learners write two sentences in their exercise books: (1) What did Akinyi actually see? (2) How could she be SURE about something she never saw? Learners then share their sentences with a neighbour before a whole-class share-out.	Teacher reads the scenario aloud slowly and with emphasis, then says: 'Before I teach you anything new today — I want your own thinking. Write two sentences only. You have 3 minutes.' [Allow 3 minutes silent individual writing.] After writing, teacher says: 'Turn to the person next to you. Share what you wrote. 1 minute.' [Allow 1 minute pair discussion.] Teacher then cold-calls exactly 4 learners (two from the left side of the room, two from the right): 'Waweru — read me your sentence 1.' 'Atieno — what did you write for sentence 2?' 'Omondi — do you agree with Atieno? Why?' 'Njeri — anything different from what we have heard?' [WAIT TIME: 10–12 seconds after each cold-call before accepting or redirecting.] Teacher does NOT correct or validate at this stage — simply says: 'Hold onto those ideas. By the end of today you will have a precise answer.' Teacher writes on the board in large letters: EVIDENCE → REASONING → INFERENCE. Teacher says: 'This is what today is about.'	Think-Pair-Share with targeted cold-calling. This activates prior intuitive knowledge from Lessons 1–4, creates cognitive dissonance (learners realise they lack precise vocabulary for what the farmer did), and primes them to receive the formal ERI framework with genuine curiosity. Writing before talking ensures all learners — not just the most vocal — commit to an answer.	Teacher listens to pair discussions and scans written sentences during the 3-minute writing window. Observable indicator: Can the learner distinguish between what Akinyi SAW (wilting, yellowing, no visible pest) and what she CONCLUDED (soil pest destroying roots)? Learners who conflate observation and conclusion are flagged for closer attention during the Explain Phase.
Observe Phase 📹 **VIDEO:** When to use z or t statistics in significance tests Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Bathymetric Evidence for	Teacher presents three scenarios on the board or reads them aloud. Learners listen and, for each scenario, copy and begin completing a three-column table in their exercise books with headers: EVIDENCE (What was observed / detected) | REASONING (The logic connecting evidence to conclusion) | INFERENCE (The conclusion reached without direct observation). Scenario A (Kisumu Farmer — already familiar): Akinyi sees wilting and yellowing along	Teacher writes all three scenario titles on the board: 'A — Kisumu Shamba, B — Nairobi Hospital, C — Amboseli Park.' Teacher says: 'These are three real types of situations in Kenya where a professional or a person reaches a conclusion about something they cannot directly see. Your job is to be a detective. For each scenario, find: What did they OBSERVE? What REASONING did they apply? What did they CONCLUDE — their INFERENCE?' Teacher	Three-Scenario Parallel Analysis. By presenting three structurally identical situations across different Kenyan professional contexts (farming, medicine, conservation), learners begin to abstract the common pattern — the ERI framework — before it is explicitly named. This is inductive sensemaking: the structure emerges from the examples rather than being imposed top-down. The table format externalises thinking and makes the distinction between	During circulation, teacher checks: (1) Is Column 1 (Evidence) filled with observable facts only — no interpretations? (2) Is Column 3 (Inference) a conclusion, not a repetition of the evidence? Key misconception to catch: learners who write 'soil pest' in Column 1 for Scenario A — this means they have not yet separated observation from inference. Teacher mentally notes these learners for targeted questioning in the Explain Phase.

Seafloor Spreading (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	one edge of the shamba; the rest of the field is healthy; she watered and fertilised evenly; no visible pest above ground; it is the dry season when soil pests are most active. Scenario B (Nairobi Radiologist): Dr. Kamau at Kenyatta National Hospital holds up a chest X-ray. He has never opened this patient's chest. He sees a white cloudy shadow on the lower right lung. He tells the nurse: 'This patient has a bacterial infection — likely pneumonia, right lower lobe.' Scenario C (Amboseli Ranger): Warden Lesiamon is patrolling Amboseli National Park at dawn. He finds: fresh footprints in the mud (size 44 boot, heading northeast), a broken acacia branch at 1.5 m height, a discarded plastic bag with a Nairobi shop label, and elephant dung that has been disturbed. He radios base: 'One poacher, adult male, passed here within the last 3 hours, heading towards the Tanzanian border.' Learners copy all three scenarios (or a shortened version dictated by teacher) and begin filling the table independently for 6 minutes.	reads each scenario slowly, pausing after each piece of evidence and asking rhetorically: 'Is that something they saw directly? Yes. Write it in column 1.' Teacher then circulates during the 6-minute individual work time, crouching beside learners to check table entries. Teacher whispers corrective prompts to struggling learners: 'That sentence you wrote — is that what he actually saw with his eyes, or is that a conclusion he jumped to?' Teacher does NOT complete the table for learners at this stage. After 6 minutes, teacher says: 'Stop writing. We will now build the correct table together.' [WAIT TIME before moving to whole-class construction: 10 seconds of silence to allow learners to re-read their own notes.]	evidence and inference visible and correctable.	
Explain Phase 📹 **VIDEO:** When to use z or t statistics in significance tests Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Bathymetric Evidence for Seafloor Spreading (Flexbook)	Teacher leads a whole-class construction of the completed ERI table on the board, taking answers from learners via cold-call and building the correct version collaboratively. After the table is complete, teacher delivers a focused 8-minute direct instruction segment formally defining the principle of inference, its components, and why it is the foundation of all scientific knowledge. Learners write the formal definition in their exercise books and annotate their tables with labels. Learners then	PHASE 3A — Table Construction (8 minutes): Teacher says: 'Let us build this table together. Scenario A first. Who can tell me ONE piece of evidence Akinyi had — something she actually observed with her senses?' Cold-calls 3 learners for Scenario A evidence column. [WAIT TIME: 12 seconds each.] For each response, teacher asks the class: 'Is that an observation or a conclusion? Thumbs up for observation, thumbs sideways for conclusion.' Teacher writes accepted answers on the board. Repeat for	Collaborative Table Construction followed by Formal Definition followed by Immediate Transfer Task (I Do – We Do – You Do). The collaborative table construction allows learners to arrive at the formal definition through their own language before the teacher names it precisely — this reduces cognitive load and increases retention. The 'Inference ≠ Guess' contrast directly addresses the most common misconception. The Kericho transfer task immediately tests whether learners can apply the	During Apply It task, teacher checks 4–5 books for the Kericho scenario. Observable success indicator: learner writes a Column 3 (Inference) entry that names something not directly observed (fungal infection) AND a Column 1 (Evidence) entry that contains only observable facts (brown curling leaves, localised to one plot). Partial understanding indicator: learner writes 'fungal infection' in both Column 1 and Column 3. No-understanding indicator: learner writes 'the leaves are sick' as the inference — this shows they have

Source: CK-12 🔍 Search ARES for similar readings	complete a brief individual 'Apply It' task: they read one new short scenario (a Kericho tea farmer notices tea leaves on one plot turning brown and curling even though rainfall has been equal — she concludes a fungal infection is present) and independently write the three ERI columns for it in 3 minutes.	Reasoning and Inference columns. Repeat the process for Scenarios B and C — cold-calling a total of 9 different learners across the three scenarios (3 per scenario). Teacher says after each completed inference column: 'Dr. Kamau has never opened that chest. Warden Lesiamon has not seen the poacher. Akinyi has not dug up the root. But they all reached a reliable conclusion. How? That is what we now formalise.' PHASE 3B — Direct Instruction on Principle of Inference (8 minutes): Teacher writes the formal definition on the board and says: 'Copy this word for word.' Definition: 'Inference is a conclusion drawn from available evidence combined with prior knowledge and logical reasoning. It is NOT based on direct observation of the thing being concluded.' Teacher then explains the three components by pointing to the board: '1 — EVIDENCE: what you can directly observe, measure, or detect. 2 — REASONING: the prior knowledge and logic that connects your evidence to a conclusion. 3 — INFERENCE: the conclusion about something you could not directly observe.' Teacher says: 'In science, most of what we KNOW came from inference. A doctor reading an X-ray. A geologist reading rock layers. A lab technologist reading a blood test result. None of them saw the disease, the ancient volcano, or the parasite directly. General Science is built on this principle.' Teacher writes on the board: 'Inference ≠ Guess. Inference = Evidence + Reasoning.' Teacher says: 'Repeat after me: Inference	framework to a new scenario without scaffolding, revealing genuine understanding versus surface copying.	not abstracted beyond description. Teacher uses cold-call responses to publicly model the correct answer for any learners still conflating observation and inference.

		is NOT a guess.' [Class repeats.] 'It is evidence plus reasoning.' [Class repeats.] PHASE 3C — Apply It Task (3 minutes): Teacher dictates the Kericho scenario and says: 'You have 3 minutes. Three columns. Go.' [Silent individual work. Teacher circulates and checks 4–5 books.] After 3 minutes: 'Pens down. Mwangi — read me your inference for the Kericho farmer.' [Cold-call 2 learners. WAIT TIME: 10 seconds each.] Teacher confirms correct answer: 'The inference is: a fungal infection is present on that plot. The evidence is: brown curling leaves on one plot only, equal rainfall. The reasoning is: fungal infections are known to cause leaf browning and curling and spread locally, not uniformly.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [When to use z or t statistics in significance tests] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Bathymetric Evidence for Seafloor Spreading (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Teacher directs learners' attention to the class Driving Question Board (DQB), which has been displayed since Lesson 1. The board carries the driving question: 'How does a scientist know something is true when they cannot see it directly — and how is this power of General Science shaping life, careers, and the environment in Kenya?' Learners are given 2 minutes to write ONE new sentence on a sticky note or piece of paper that they will contribute to the board. The sentence must complete one of two stems (teacher writes both on the board): Stem 1: 'A scientist can know something without seeing it because…' Stem 2: 'I now understand that the Kisumu farmer knew about the soil pest because…' Learners post or display their sentences. Teacher reads 3–4 aloud and annotates the	Teacher points to the DQB and says: 'This board has been asking us a big question since Lesson 1. Today we have a new answer to add. Not a guess — a scientific answer.' Teacher writes both sentence stems on the board and says: 'Choose one stem. Write one sentence only. Make it precise. You have 2 minutes.' [Allow 2 minutes silent individual writing. No pair discussion at this stage — this is an individual consolidation moment.] Teacher then collects or invites learners to read aloud 3–4 sentences. For each, teacher says: 'Is that an inference or is that just a description? Let us check.' Teacher annotates the DQB with a large arrow connecting 'How does a scientist know?' to the new label 'INFERENCE = Evidence + Reasoning (Lesson 5).' Teacher says: 'The Kisumu farmer is no longer mysterious to us. She used	Driving Question Board Sentence-Stem Contribution. The sentence stem lowers the barrier to participation while requiring learners to synthesise today's learning in their own words and connect it explicitly to the anchoring phenomenon. Publicly posting and reading the sentences creates collective accountability and reinforces the idea that scientific understanding is cumulative — each lesson adds a new layer to the same central question.	Teacher reads the posted sentences and checks: (1) Does the sentence use the word 'inference', 'evidence', or 'reasoning'? (2) Does the sentence correctly link the Kisumu farmer's conclusion to indirect evidence? Sentences that say 'she knew because she is a good farmer' reveal that the learner has not yet internalised the formal principle. These learners are noted for follow-up in Lesson 6.

	DQB with a new label: 'INFERENCE — Lesson 5 unlocked.'	the principle of inference — evidence plus reasoning. And so does every scientist in Kenya and around the world.'		
Model Building Phase 📹 **VIDEO:** [When to use z or t statistics in significance tests] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Bathymetric Evidence for Seafloor Spreading (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners review their completed three-scenario ERI table and add a fourth row at the bottom: 'MY OWN EXAMPLE.' They must write one real-life Kenyan example of inference from their own experience or general knowledge — not from the three scenarios taught today. Examples might include: a parent inferring a child has malaria from fever and shivering; a bus driver in Nairobi inferring the engine has a problem from a strange sound; a fisherwoman on Lake Victoria inferring a storm is coming from the colour of the sky. Learners then complete a 3-question exit card on a small piece of paper or in their exercise books.	Teacher says: 'Your ERI table is now a scientific model. It shows you the structure of all scientific reasoning. I want you to own this model by adding one row from YOUR life in Kenya. Think of a time — at home, on the shamba, on the road, at the market — when someone reached a conclusion about something they could not see directly. You have 2 minutes.' [Allow 2 minutes individual silent work. Teacher circulates and reads 4–5 examples, nodding affirmatively at correct ones and whispering 'Is that observation or inference?' at incorrect ones.] After 2 minutes, teacher cold-calls 2 learners to share their examples: 'Auma — what did you write?' 'Kipchoge — yours?' [WAIT TIME: 10 seconds each. Teacher validates: 'That is a genuine inference — evidence plus reasoning, conclusion about something not directly seen. Well done.'] Teacher then says: 'Last task. Exit card. Three questions. Answer in your exercise book. You have 3 minutes.' Teacher writes the three questions on the board: Q1: Write the definition of inference in your own words. Q2: In the Amboseli ranger scenario, name ONE piece of evidence and state the inference Warden Lesiamon made. Q3: Why is inference described as the foundation of all scientific knowledge? Give one reason. [3 minutes silent individual writing. Collect or check books at the door as learners leave, or during the	Own-Example Generation + Exit Card. Asking learners to generate their own Kenyan inference example is a high-order transfer task (application to novel context) that consolidates abstract understanding. The exit card serves a dual purpose: it is both a formative assessment tool for the teacher and a self-assessment moment for the learner. The three-question design moves from recall (Q1 — definition) to application (Q2 — specific scenario) to analysis (Q3 — why inference is foundational), mirroring the cognitive progression of the lesson as a whole.	Exit card Q1 success indicator: learner definition must include 'evidence', 'reasoning', and 'not directly observed' (or equivalent phrasing). Q2 success indicator: evidence is a specific observable fact (e.g., 'size 44 boot footprints in the mud') and inference names something not seen (e.g., 'one adult male poacher passed within 3 hours'). Q3 success indicator: learner articulates that most scientific knowledge is about things too small, too far, too past, or too hidden to observe directly — therefore inference is the universal method. Exit cards with vague or circular definitions (e.g., 'inference is when you infer something') flag learners requiring re-teaching at the start of Lesson 6.

		next lesson's opening.]		

D. TEACHER REFLECTION
After the lesson, consider the following: (1) How many learners in the Apply It task (Kericho scenario) correctly separated evidence from inference on first attempt — and what does this tell you about the readiness of the class for Lesson 6, which will require learners to design their own inference-based investigations? (2) Were the three Kenyan scenarios (farmer, doctor, ranger) sufficiently familiar to all learners, or did learners in the class who have never been to a hospital or Amboseli struggle to contextualise Scenarios B and C? If so, consider substituting Scenario B with a mother in Kibera inferring her child has worms from persistent stomach pain and weight loss, and Scenario C with a Mombasa fisherman inferring a school of tilapia is nearby from the circling of birds above the water. (3) Did the 'Inference ≠ Guess' statement land clearly? If learners are still using 'guess' and 'inference' interchangeably in exit cards, open Lesson 6 with a 5-minute contrast exercise: show two scenarios side by side — one where a conclusion is made with no evidence (a guess), one where it is made with structured evidence and reasoning (an inference) — and ask learners to vote which is which. (4) Review all exit cards before Lesson 6. Group them into three piles — clear understanding, partial understanding, and significant confusion — and plan one targeted re-engagement question at the start of Lesson 6 for the partial/confusion groups.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	In three Kenyan scenarios — the Kisumu maize farmer, the Nairobi radiologist Dr. Kamau at Kenyatta National Hospital, and Amboseli Warden Lesiamon — each professional reached a reliable conclusion about something they could NOT directly see: a soil pest destroying roots, a bacterial infection in a patient's lung, and a poacher's path through the park. Learners observed that in every case, the conclusion was reached by systematically examining available physical evidence.
What did I learn?	Inference is a conclusion drawn from available evidence combined with prior knowledge and logical reasoning — it is NOT based on direct observation of the thing being concluded, and it is NOT a guess. Scientific inference has three components: (1) Evidence — what can be directly observed or detected; (2) Reasoning — the prior knowledge and logic connecting evidence to a conclusion; (3) Inference — the conclusion about something not directly observable. The principle of inference is the foundation of all scientific knowledge because most scientific phenomena (soil pests, diseases, subatomic particles, ancient events) cannot be directly observed.
How does this explain the phenomenon?	The Kisumu farmer Akinyi knew there was a soil pest destroying her maize roots — even though she never saw the pest — because she applied the principle of inference: her evidence (wilting and yellowing confined to one edge, uniform watering and fertilising, dry season timing, no visible above-ground pest) combined with her reasoning (soil pests are seasonally active in dry periods and cause localised root damage that produces exactly this pattern of wilting) led her to a reliable inference (a soil pest is destroying the roots underground). This is not a guess — it is science.

LESSON 6 (80 minutes): Collecting Evidence and Drawing Inferences: Mystery Bags Hands-On Activity

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To apply scientific evidence-collection methods and inference skills through a structured hands-on group activity, consolidating the principle that reliable scientific conclusions can be drawn from indirect evidence.
Knowledge	Learners will be able to: (a) identify the methods scientists use to collect evidence (observation, measurement, experimentation, record-keeping); (b) map evidence-collection methods onto the KICD science process skills — classifying, measuring, hypothesising, identifying variables, and interpreting data; (c) distinguish between an observation and an inference; (d) explain how scientists analyse indirect evidence to arrive at reliable conclusions.
Skills	Classifying written clues; hypothesising a hidden cause from pattern evidence; identifying variables within a set of clues; interpreting data (written descriptions) to construct an inference; communicating and justifying an inference to peers; designing and evaluating conclusions.
Attitudes	Appreciate that rigorous evidence-collection — not guessing — is what makes scientific inference trustworthy; respect peers' competing inferences and evaluate them against evidence rather than dismissing them; develop confidence in using systematic reasoning even when direct observation is impossible.
Key Inquiry Question	How is General Science useful in daily life?
Purpose in Storyline	In Lessons 1–5, learners encountered the anchoring phenomenon (the Kisumu maize farmer inferring a soil pest from wilting patterns), explored the meaning of General Science, its importance in Kenyan life and careers, and the formal principle of inference. Lesson 6 is the first full hands-on application lesson. Learners now embody the role of the scientist: they receive only written evidence clues (no images, no direct observation) and must construct an inference, justify it, and defend it publicly — mirroring exactly what the farmer did. This lesson advances the Driving Question Board by moving from 'What is inference?' to 'How do we DO inference systematically?', and directly demonstrates that evidence-collection methods are the tools that make indirect inference reliable.
Safety Notes	No hazardous materials are used. All mystery bag contents are sealed written cards — nothing edible, sharp, or chemical. Remind learners not to open the sealed envelopes. Learners with visual impairments should be seated so a peer can read clue cards aloud. Ensure equitable group roles so no single learner dominates the evidence-reading process.

B. LESSON OVERVIEW

Learners work in groups of 4–5 with sealed 'mystery evidence bags' — envelopes containing written clue cards describing a plant, its soil, smell, leaf texture, growth pattern, and surrounding environment in a Kenyan farming context (no images). Each group must infer the hidden cause (a plant disease, nutrient deficiency, or soil pest) using only written evidence. Groups then present their inferences to the class, justify each claim with specific clue evidence, and listen to competing inferences from other groups. The teacher facilitates a whole-class discussion that maps what the groups just did onto the formal KICD science process skills: classifying, measuring, hypothesising, identifying variables, and interpreting data. The lesson closes by connecting the activity back to the anchoring phenomenon — the Kisumu farmer — and students update the Driving Question Board with new 'How do scientists…?' questions generated by the activity.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [When to use z or t statistics in significance tests] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Scale and Indirect Measurement Applications (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Before any mystery bags are distributed, each learner silently reads a short scenario card placed on their desk: 'A KEPHIS plant health inspector in Nakuru has been called to a flower farm. She cannot enter the greenhouse yet — biosecurity protocols prevent her from touching the plants. Standing outside, she can only read a written field report from the farm worker describing: leaf colour changes, soil moisture level, the smell near the roots, patterns of which rows are affected, and the time of year the symptoms started. From this written report alone, she must decide whether to quarantine the farm. What would you do? What clues would matter most to you? Write down: (1) your best guess at what the problem might be, and (2) two or three specific pieces of evidence from the report description you think would be most important.' Learners write individually for 4 minutes. Teacher then calls on 3 learners using cold-call (not volunteers) to share their top evidence clue — one sentence only. Teacher records the three named clues on the board under the heading: 'Clues we think matter.'	Distribute scenario cards face-down. Say: 'Do not turn over until I say go.' After 30 seconds of settling: 'Turn over and read silently. You have 4 minutes to write your two answers. No talking yet.' Set a visible timer. At 3 minutes: 'One more minute — make sure you have written at least one specific clue.' At 4 minutes: 'Pens down.' Cold-call three learners by name (choose one high-confidence, one mid-confidence, one quieter learner identified from prior lessons): 'Amina — in one sentence, what is the single most important clue? Wait 10 seconds.' Record each response verbatim on the board. After all three: 'Notice that we already disagree about which clues matter most. That disagreement is the heart of today's lesson. Keep your prediction card — you will need it at the end.'	Individual written prediction (Think-Before-You-Act): By committing a prediction to paper before any activity, learners activate prior conceptions about evidence and inference, creating a cognitive anchor that the Observe phase will either confirm or productively disrupt. This mirrors the farmer's mental process: she too formed a prior hypothesis before systematically checking each clue.	Teacher scans written prediction cards as learners write (circulate for 2 minutes). Observable indicator: learner has written at least one specific clue (not just 'the plant looks sick') and one named possible cause. Cards are collected at the end of the lesson to check for conceptual shift between prediction and final explanation.
Observe Phase 📹 **VIDEO:** [When to use z or t statistics in significance tests] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Groups of 4–5 receive one sealed mystery evidence bag each. There are five different bags in total (A–E), each representing a different Kenyan farming scenario. Bag A: Maize plants near a Kisumu riverbank — clues describe yellowing lower leaves, soft mushy stem base, a faint sour smell in the soil, waterlogged soil texture, no visible insects, symptoms only in	Distribute bags. Say: 'Do NOT open the bag yet. First, assign your roles — you have 1 minute.' Circulate to confirm roles are assigned. Then: 'Open the bag. You have 18 minutes. Start by having your Reader read each card aloud to the whole group before anyone writes anything.' At 8 minutes, give a mid-point check: circulate and ask each group's	Evidence Classification Table (Structured Data Sorting): Providing a three-column classification table (pest / disease / nutrient-environment) forces learners to sort evidence rather than jump to conclusions — this is the science process skill of classifying made explicit and tactile. It also surfaces contradictory clues, which is the	Circulate and inspect Evidence Classification Tables. Observable indicators: (1) group has classified at least 5 of the available clues; (2) group inference sentence names a specific cause (not just 'something is wrong with the plant'); (3) group has identified at least one 'confusing' or 'contradictory' clue — this indicates deeper engagement with evidence quality,

📖 **HTML:** [Scale and Indirect Measurement Applications (Flexbook)] Source: CK-12 🔍 [Search ARES for similar readings]	the lowest-lying rows. Bag B: Bean plants on a Rift Valley smallholding — clues describe purple-tinged leaves, stunted growth, soil described as 'pale and sandy,' no pest tracks visible, farmer recently cleared the land and has not added fertiliser. Bag C: Tomato seedlings in a Kiambu greenhouse — clues describe brown circular spots on leaves, white powdery coating on some leaves, high humidity noted in the report, neighbouring greenhouse had similar symptoms two weeks ago, no pest insects found. Bag D: Sukuma wiki (kale) in a Mombasa coastal shamba — clues describe small holes in leaves that are not visible at the time of inspection, but the farm report notes the holes appeared overnight, droppings found on the soil surface, symptoms appear in patches, not uniform rows. Bag E: Sorghum on a Baringo farm — clues describe stunted plants with swollen, distorted stems, a sticky substance on the stem surface, the distorted plants are clustered in the centre of the field, surrounding plants appear normal. Each group reads all clue cards aloud (roles: Reader, Recorder, Evidence Sorter, Timekeeper, Presenter). Groups must: (1) list every piece of evidence they have been given; (2) classify each clue as 'Points towards a pest,' 'Points towards a disease,' or 'Points towards a nutrient/environmental problem' using a provided Evidence Classification Table; (3) record their group inference — one sentence — and identify the two strongest pieces of evidence supporting it; (4) note any clues they found confusing or	Recorder: 'How many clues have you classified so far?' If a group is stuck on classification, prompt: 'Is this clue something you can observe directly, or is it something someone measured or recorded over time? That distinction will help you.' At 15 minutes: 'You should be writing your group inference sentence now — one sentence, naming the cause AND the two best pieces of evidence.' At 18 minutes: 'Pens down. Presenter — practise your one-sentence inference in your group for 2 minutes before we share.' Do NOT confirm or deny any group's inference at this stage. If asked 'Are we right?', respond: 'Hold that question — the evidence will tell us. What does your strongest clue say?'	most important cognitive moment: learners discover that real scientific evidence is rarely perfectly consistent, and inference requires weighing evidence, not just collecting it. This directly mirrors how the Kisumu farmer weighed the absence of visible insects against the pattern of root damage.	which is the target process skill for this lesson.

	contradictory.			
Explain Phase 📹 **VIDEO:** When to use z or t statistics in significance tests Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Scale and Indirect Measurement Applications (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each group's Presenter shares their group's inference to the class in 2 minutes maximum, using the sentence frame displayed on the board: 'Based on [evidence clue 1] and [evidence clue 2], our group infers that the cause is [named cause], because [reasoning that connects the clues to the cause].' After all five groups present, the teacher facilitates a structured whole-class discussion: (1) Were any two groups working on the same bag? (If so, did they reach the same inference? Why or why not?) (2) Which group had the most contradictory clues — and how did they handle them? (3) The teacher then formally names and maps what the class just did onto the KICD science process skills: 'What you called sorting clues — scientists call classifying. What you called your best guess — scientists call hypothesising. What you called the strongest clues — scientists call identifying key variables. What you wrote in your table — scientists call recording and interpreting data. And what you just told the class — that is drawing and communicating a conclusion.' Teacher writes this mapping on the board as a two-column table: 'What WE did in the activity' | 'What scientists call it.' Learners copy this into their science notebooks. Teacher then returns to the anchoring phenomenon: 'The maize farmer in Kisumu did every single one of these steps — without a lab, without equipment, just from patterns she observed over time. Which of these process skills do you think she used most? Discuss	Manage presentations with a visible 2-minute timer per group. After each presentation, ask the class (not the presenting group): 'What is the one piece of evidence from that group's bag that you find most convincing? Raise your hand.' Take 2 responses — cold-call one, take one volunteer. After all five presentations, draw the two-column mapping table on the board and build it collaboratively: 'Tell me — what did you physically DO in step one of the activity?' Accept responses, then write the scientific name. Do not give the scientific names first — elicit the student action first. For the farmer connection: 'Turn to your partner. 45 seconds — which process skill did the Kisumu farmer rely on most heavily, and why?' After pairs discuss: cold-call 2 pairs. If a learner says 'She classified,' ask: 'What exactly was she classifying?' WAIT 10 seconds before accepting an answer. End the explain phase by saying: 'So inference is not magic — it is a structured process. That is what makes it science and not just guessing.'	Structured Academic Controversy with Sentence Frames: The 2-minute presentation with a mandatory sentence frame ('Based on… our group infers… because…') requires learners to construct a claim-evidence-reasoning chain — the core structure of scientific explanation. The post-presentation whole-class mapping exercise (student action → scientific term) builds scientific vocabulary through experience-first naming, which research shows produces deeper retention than definition-first instruction. Connecting back to the farmer at the close anchors all new vocabulary in the phenomenon.	Listen for correct use of the sentence frame. Observable indicators: (1) presenter names a specific cause (not vague); (2) presenter cites at least two specific clues from their bag (not general statements); (3) during the farmer-connection pair discussion, learners name at least one specific process skill by its scientific name. Teacher notes which groups struggled to identify 'contradictory clues' during the Observe phase — these groups receive a targeted follow-up question during the DQB phase.

	with your partner for 1 minute.' Cold-call 2 pairs for responses.			
Driving Question Board (DQB) Creation 📹 **VIDEO:** [When to use z or t statistics in significance tests] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Scale and Indirect Measurement Applications (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each learner receives two sticky notes. On the first (yellow): write one new question that today's activity raised for you about how scientists collect or evaluate evidence — a question you did not have before this lesson. On the second (green): write one thing you now understand about the Kisumu farmer's inference that you did not fully understand before today. Learners post both sticky notes on the class Driving Question Board under two new columns added today: 'New Questions About Evidence' and 'Growing Understanding of the Farmer.' The teacher reads aloud 4–5 of the yellow sticky notes and groups them into clusters on the board (e.g., questions about measurement, questions about when evidence is 'enough,' questions about peer review). Teacher says: 'Look at these clusters. These are exactly the questions that professional scientists in Kenya — agronomists at KARI in Kikuyu, epidemiologists at KEMRI in Nairobi, environmental scientists monitoring Lake Victoria — wrestle with every day. Your questions are real scientific questions.' Teacher points to the growing DQB and connects it to the sub-strand driving question: 'Our driving question asks how a scientist knows something is true when they cannot see it directly. Based on today, what part of that answer do we now have?'	Distribute sticky notes before the Explain phase closes so no time is lost. Give clear instructions: 'Yellow = a NEW question today gave you. Green = something you NOW understand about the farmer. 3 minutes, write silently.' Circulate and prompt any learner who is stuck on the yellow note: 'What confused you today? What would you need to know to feel MORE confident in your inference?' After posting, do not read every note — select 4–5 that represent distinct question types. Group them physically on the board using a marker to draw circles around clusters. Narrate the clustering process aloud: 'These three questions are all asking about the same thing — how much evidence is enough. Let's label this cluster: Evidence Sufficiency. We will return to this in Lesson 7.'	Two-Colour Sticky Note DQB Protocol (Questions + Understanding): Separating new questions (yellow) from growing understanding (green) prevents the common DQB problem where learners only post what they know, not what they are still puzzling over. The clustering move by the teacher makes the DQB a living, structured map of collective understanding rather than a passive display. Naming the clusters with scientific language (e.g., 'Evidence Sufficiency') plants vocabulary that will be formalised in later lessons.	Quality of yellow sticky notes is the key observable indicator. Target: learner's new question is specific to evidence or inference (not a general curiosity question unrelated to the sub-strand). Teacher photographs the DQB at the end of class for tracking across the 8-lesson sequence. Green sticky notes are checked for conceptual accuracy — any green note that reveals a persistent misconception (e.g., 'I now understand that the farmer just guessed') is addressed at the start of Lesson 7.
Model Building	Learners open their science notebooks to their running	Say: 'Open your notebooks to your Inference Model. You are adding	Cumulative Model Revision (Running Inference Model): The	Observable indicators in notebook models: (1) at least 3 evidence-

Phase 📹 **VIDEO:** When to use z or t statistics in significance tests Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Scale and Indirect Measurement Applications (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	'Inference Model' — a diagram they have been building since Lesson 3. Today they add a new layer. The current model has: Phenomenon (wilting maize) → Hidden Cause (soil pest) → Inference Arrow. Today learners must add: (1) a box labelled 'Evidence Collection Methods' branching off the inference arrow, listing the methods the farmer used (observation of patterns, record-keeping of affected rows, elimination of other causes, noting season and location) and mapping each to its KICD process skill name; (2) an annotation explaining why MORE methods = MORE reliable inference; (3) a second small diagram showing their own mystery bag: their evidence clues → their inference → the process skills they used. Learners who finish early add a third element: a 'Confidence Scale' (1–5) rating how confident they are in their inference and one sentence explaining what additional evidence would increase their confidence to a 5. At the close, 2 learners share their updated model with a partner and give one piece of feedback using the stem: 'Your model clearly shows… One thing you could add is…'	two things today — I will show you the structure on the board.' Draw a quick skeleton of the two additions on the board (box 1: Evidence Collection Methods with KICD skill names; box 2: mini mystery-bag diagram). Say: 'You have 10 minutes to add both. Start with whichever one feels clearer to you.' At 7 minutes: 'If you have both boxes, add the Confidence Scale — rate your mystery bag inference 1 to 5 and explain what evidence would make it a 5.' At 10 minutes: 'Turn to the person next to you. 2 minutes — share your model and give them ONE piece of feedback using the sentence on the board.' At 12 minutes: 'Return to your own model. Do you want to revise anything based on your partner's feedback? 1 minute.' Collect 3–4 notebooks (rotating across lessons) to review model quality. Return with written feedback by next lesson.	running model is the lesson sequence's primary sensemaking artifact. Each lesson adds a new explanatory layer, so learners can physically see their understanding deepen across the sub-strand. Today's additions are the most procedurally concrete yet — mapping their own lived activity onto the model makes the science process skills personally meaningful rather than abstract. The Confidence Scale addition introduces the scientific concept of evidence sufficiency without using that term, preparing learners for Lesson 7's formal treatment.	collection methods named and correctly mapped to KICD process skill names; (2) mystery-bag mini-diagram includes evidence clues → inference → named process skills (not just 'we sorted things'); (3) peer feedback exchange is on-task and uses the sentence stem. Teacher uses the 3–4 collected notebooks to assess whether learners are building a coherent cumulative model or treating each lesson as isolated — this informs Lesson 7 scaffolding.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record brief notes in your professional journal:

1. EVIDENCE QUALITY: When groups presented their inferences, did learners cite specific clues or did they speak in generalities ('the plant looked bad')? If generalities dominated, consider adding a 'Quote the Clue' rule in Lesson 7 — presenters must read the exact clue card sentence before making a claim.

2. CONTRADICTORY CLUES: Which group(s) identified contradictory or confusing clues in their bag? How did they handle the contradiction — did they ignore it, or did they attempt to reason through it? Note these groups for targeted questioning in Lesson 7, as reasoning through contradictory evidence is the most sophisticated

inference skill in this sub-strand.

3. PROCESS SKILL MAPPING: During the Explain phase, were learners able to match their own actions to the KICD process skill names (classifying, hypothesising, identifying variables, interpreting data)? If the mapping felt mechanical, plan a 5-minute warm-up at the start of Lesson 7 where learners try to recall the mapping from memory before it is displayed.

4. FARMER CONNECTION: When learners discussed which process skill the Kisumu farmer used most, what did the cold-called pairs say? Did any learner make an unexpected connection (e.g., linking to a farmer they know personally, or to a news story about crop disease in Kenya)? Record these connections — they are high-value anchors for Lesson 8's culminating activity.

5. DRIVING QUESTION BOARD: Review the yellow sticky notes photographed at the end of class. Are the questions becoming more precise and scientific across the lesson sequence (evidence of growing scientific thinking), or are they still broad and non-specific? If broad, plan an explicit 'question refinement' move in the next DQB update.

6. EQUITY CHECK: Did any group have a dominant member who read all clues and directed all decisions? If so, consider assigning the role of 'Evidence Challenger' in Lesson 7 — one person whose job is to ask 'What evidence are we ignoring?' before the group finalises its inference.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners opened sealed mystery evidence bags containing written clue cards describing plants in Kenyan farming contexts (Kisumu, Rift Valley, Kiambu, Mombasa, Baringo). They read, sorted, and classified written evidence clues — no images, no direct plant observation — into categories (pest / disease / nutrient-environmental problem) using an Evidence Classification Table.
What did I learn?	Learners identified and named the methods scientists use to collect evidence (observation, measurement, experimentation, record-keeping) and mapped their own group actions onto the KICD science process skills: classifying, hypothesising, identifying variables, interpreting data, and drawing conclusions. They distinguished between an observation and an inference, and recognised that contradictory evidence must be weighed — not ignored — for an inference to be reliable.
How does this explain the phenomenon?	Learners explained, using the sentence frame 'Based on [evidence] and [evidence], our group infers [cause] because [reasoning],' why indirect evidence — systematically collected and classified — is sufficient for a scientist to draw a reliable conclusion. They connected this to the anchoring phenomenon: the Kisumu maize farmer used the same evidence-collection and inference process they practised today, demonstrating that General Science process skills are tools used in real Kenyan agricultural and professional contexts every day.

LESSON 7 (80 minutes): DQB Curation and Cumulative Model Building	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To synthesise all evidence gathered across Lessons 1–6 by curating the Driving Question Board and building a personal Cumulative Science Model that connects General Science to the anchoring phenomenon, to the learner's chosen career, and to real Kenyan contexts.
Knowledge	Learners demonstrate understanding of: (a) the meaning and importance of General Science; (b) the principle of inference as used in science education; (c) career opportunities related to General Science; and (d) how scientific knowledge connects across biological, physical, and chemical phenomena.
Skills	Learners practise: classifying questions by evidence weight; writing evidence-based explanations; constructing mind-maps/diagrams that synthesise cross-lesson learning; and peer-reviewing models using assessment rubric descriptors.
Attitudes	Learners appreciate the importance of General Science in day-to-day life; demonstrate respect for peers' career choices and reasoning; and show responsibility in honest self-assessment of their own understanding.
Key Inquiry Question	How is General Science useful in daily life?
Purpose in Storyline	This is the penultimate lesson in the evidence-gathering sequence. Learners have now accumulated six lessons of evidence about how scientists make reliable conclusions from indirect evidence — anchored on the Kisumu maize farmer who inferred a hidden soil pest. Lesson 7 is the consolidation pivot: learners curate their Driving Question Board to audit what they now know, what they partly understand, and what still needs investigation. They then encode this understanding into a personal Cumulative Science Model — a mind-map or annotated diagram — that makes their thinking visible, career-connected, and tethered to the anchoring phenomenon. This sets the class up for Lesson 8's summative showcase.
Safety Notes	No hazardous materials are used. Learners use pens, paper, printed or digital mind-mapping tools. If digital devices are used, remind learners to observe responsible and safe online behaviour (cyber security PCI). Ensure peer-review conversations are conducted respectfully — the teacher should model constructive feedback language before peer-review begins.

B. LESSON OVERVIEW

Lesson 7 is a 80-minute synthesis and model-building session. The class begins by returning to the full Driving Question Board (DQB) generated across Lessons 1–6 and physically sorting every question card into one of three columns: 'We Can Now Answer This,' 'We Partly Understand This,' and 'We Still Don't Know This.' For questions in the first column, learner pairs draft concise, evidence-based explanations (1–2 sentences each) and share these with the class, anchoring every explanation back to the Kisumu maize farmer phenomenon. The teacher then delivers a brief, forward-looking 'storyline bridge' that maps unanswered DQB questions onto future sub-strands (The Cell, Nutrition, Respiration, etc.), showing learners that their curiosity is the engine of the full Grade 10 course. In the second half, each learner constructs or substantially updates their personal Cumulative Science Model — a mind-map or annotated diagram — that must include: (1) the anchoring phenomenon (the Kisumu farmer's inference), (2) the principle of inference, (3) at least two real Kenyan career or life contexts, and (4) connections to at least two other science topics encountered in the unit. The lesson closes with a structured peer-review using KICD rubric descriptors, followed by a whole-class gallery reflection.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Analyzing a cumulative relative frequency graph] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cumulative Frequencies (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives a blank 'Knowledge Audit Card' (a half-sheet of paper divided into three columns: 'I can explain this with evidence,' 'I partly understand this,' 'I am still confused about this'). Working individually for 5 minutes, learners silently scan the DQB (all question cards from Lessons 1–6, still posted on the wall or projected) and write at least one question from the board into each of their three columns. They then write a one-sentence prediction: 'By the end of this lesson, I expect to be able to answer ___ of our DQB questions fully.' Learners keep their audit cards — they will revisit them at the lesson's close. This activates metacognitive self-monitoring and creates personal stakes in the curation exercise.	Before learners begin, teacher says: 'Look at every question on our DQB. These came from YOUR minds across six lessons. Today we are going to act like scientists reviewing a research log — sorting what we know from what we still need to investigate.' Teacher circulates silently for the full 5 minutes, noting 2–3 audit cards that show interesting or surprising self-assessments (these will be used as cold-call prompts later). After 5 minutes, teacher cold-calls 3 learners (spread across the room): 'Amina, read me one question you placed in your 'I can explain' column and tell us in one sentence why you feel confident.' WAIT TIME: 12 seconds after each cold-call before accepting the response. Teacher does NOT correct or confirm at this stage — respond with: 'Hold that thought — let's see if the class agrees when we curate the board together.'	Metacognitive Knowledge Audit: By forcing learners to categorise their own understanding before instruction, this strategy surfaces misconceptions and genuine confidence, making the subsequent DQB curation feel personally meaningful rather than a teacher-directed sorting exercise. It also operationalises the NGSS Science and Engineering Practice of 'Obtaining, Evaluating, and Communicating Information' at a reflective level.	Observable indicator: Each learner has at least one question in each of the three columns of their audit card. Teacher scans cards during circulation — flag any learner who has placed ALL questions in 'I can explain' (possible overconfidence) or ALL in 'I am still confused' (possible disengagement) for targeted support during the curation phase. Collect 4–5 audit cards to inform pacing.
Observe Phase 📹 **VIDEO:** Analyzing a cumulative relative frequency graph Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cumulative Frequencies (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class works as a whole group (with learners in their established lab groups from previous lessons) to physically curate the DQB. The teacher reveals three large labelled columns on the board or wall: Column A — 'We Can Now Answer This (with evidence from Lessons 1–6),' Column B — 'We Partly Understand This (we have some evidence but gaps remain),' Column C — 'We Still Don't Know This (we need more science).' Each lab group is assigned 3–4 DQB question cards. Groups spend 8 minutes discussing which column their assigned questions belong in, writing a brief justification on the back of each	Teacher facilitates rather than directs. As groups place cards, teacher asks targeted probing questions: 'You placed that in Column B — what specific piece of evidence from Lesson 3 would need to be stronger before you'd move it to Column A?' WAIT TIME: 15 seconds. Cold-call 4 different learners across the curation (rotate groups). For each Column A answer drafted on a sticky note, teacher checks the phenomenon connection by asking: 'Before we accept this answer, can someone show me how this connects back to Mama Akinyi's maize shamba in Kisumu?' (Use the farmer's name consistently to maintain narrative	Collaborative Evidence Sorting (Argument-Driven Inquiry): Requiring groups to justify placements with specific lesson-derived evidence enacts the NGSS practice of 'Engaging in Argument from Evidence.' The public, challengeable nature of the sorting ensures that explanations are held to an evidential standard, not just opinion. Anchoring all Column A answers back to the phenomenon ensures that sensemaking is cumulative and narrative, not fragmented.	Observable indicator: Each group correctly justifies the placement of at least 2 of their 3–4 cards using specific evidence from named lessons. Teacher listens for vague justifications (e.g., 'We know this because we learned it') and redirects: 'Which specific observation or data from which lesson?' A question is only moved to Column A if the class can produce a sticky-note answer that references evidence AND the phenomenon. Teacher records which DQB questions generate the most disagreement — these are priority topics for Lesson 8 review.

	card ('We can answer this because in Lesson [X] we found that…'). Groups then take turns placing their cards on the board and reading their justification aloud. The class may challenge a placement — disputes are resolved by asking: 'What specific evidence from our lessons supports this answer?' For questions in Column A, a volunteer pair drafts a 1–2 sentence evidence-based answer on a sticky note and attaches it below the question card. The teacher ensures that at least 3 answered questions are explicitly linked back to the anchoring phenomenon: 'How does this answer help us understand how the Kisumu farmer knew there was a soil pest underground?' This phase takes approximately 20 minutes.	coherence.) If a connection is weak, teacher models the bridging: 'Notice how inference — drawing a conclusion from indirect evidence — is exactly what this answer demonstrates. That is the same move the farmer made.' Teacher also narrates the Column C questions with forward momentum: 'These questions in Column C are not failures — they are your research agenda. We will mark them with a star because they are the questions that will take us into the next sub-strands.' Write on the board: 'Our unanswered questions will be answered by: The Cell (1.2), Nutrition (1.3), Respiration (1.5)…' — mapping 2–3 specific unanswered DQB questions onto specific future sub-strands by name.		
Explain Phase 📹 **VIDEO:** Analyzing a cumulative relative frequency graph Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cumulative Frequencies (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher delivers a 10-minute structured 'Storyline Bridge' talk — not a lecture, but a guided narration. Learners have their science notebooks open and draw a simple timeline of the Grade 10 General Science course (sub-strands 1.1 through 3.4) as the teacher speaks. The teacher maps each major unanswered DQB question onto a future sub-strand, showing learners that their curiosity from the Kisumu farmer phenomenon threads through the entire course. For example: 'You asked, How do plants know they are being attacked by pests underground? — that question will get a much richer answer in Sub-Strand 1.4 Transport in Plants and Sub-Strand 1.6 Plant Growth and Development.' After the bridge talk, learners individually complete a 'Career–Science–Phenomenon	During the storyline bridge, teacher uses the course summary table in the KICD curriculum design as a visual (projected or drawn on board), pointing to each sub-strand and saying: 'This is where your question about X will find its answer.' Maintain the farmer narrative: 'Mama Akinyi's question — how do I know something is real if I cannot see it — is a question that every doctor reading an X-ray in Kenyatta National Hospital asks, every lab technologist testing blood at Aga Khan Hospital asks, and every environmental scientist measuring pollution at Lake Victoria asks.' After career triangles are drawn, cold-call 2 volunteers (ensure gender balance): 'Otieno, show us your triangle and read us your three connecting sentences.' WAIT TIME: 12 seconds between prompt	Storyline Bridging and Career Triangulation: The storyline bridge enacts the NGSS crosscutting concept of 'Patterns' at the course level — showing learners that the pattern of inference repeats across all scientific disciplines. The Career–Science–Phenomenon Triangle operationalises KICD's core competency of Self-Efficacy by requiring learners to place themselves inside the scientific story, not just observe it from outside.	Observable indicator: Each learner's Career–Science–Phenomenon Triangle has three labelled nodes AND three labelled arrows, each arrow carrying a complete sentence. Teacher scans notebooks during the sharing phase — flag any learner whose arrows are unlabelled or whose career connection is not science-specific (e.g., 'I want to be a singer' without a science link) for a brief one-on-one nudge: 'Think about how a sound engineer or a music producer uses science — inference is everywhere.'

	Triangle' in their notebooks: a three-point diagram where Point 1 is their chosen career (e.g., agronomist, doctor, environmental scientist, lab technologist, engineer), Point 2 is the principle of inference, and Point 3 is the Kisumu maize farmer phenomenon. Learners draw connecting arrows and label each arrow with one sentence explaining the connection. Two volunteers share their triangles with the class. This prepares learners for the model-building phase.	and response. After each sharing, teacher asks the class: 'Does anyone have a different career that also relies on inference? Raise your hand.' Take 2–3 hands. Teacher affirms: 'Notice that almost every career in Kenya that depends on science uses inference — that is the power we have been studying for six lessons.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Analyzing a cumulative relative frequency graph Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Cumulative Frequencies (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	With the three-column DQB now fully curated, learners conduct a 10-minute 'DQB Final Review' in pairs. Each pair is given a colour-coded dot sticker: GREEN for Column A questions they feel the class explained well, YELLOW for Column B questions they want to revisit in Lesson 8, and RED for Column C questions they are most curious about for future sub-strands. Pairs place their dots on the relevant question cards, then each pair writes one new question on a blank card — a question that emerged for them during today's curation that was NOT already on the board. These new cards are added to Column B or C. The class reads the new additions aloud. The teacher records the final DQB state (photograph or written list) as the official class record of their growing understanding. The teacher then narrates the DQB as a living scientific document: 'Real scientists at the Kenya Medical Research Institute (KEMRI) maintain boards exactly like this — they call them research gaps. Your	Distribute dot stickers before the pair activity begins. As pairs place dots, teacher circulates and asks: 'Why yellow and not green for this one — what evidence is still missing?' WAIT TIME: 10 seconds. After new cards are added, teacher cold-calls 3 pairs to read their new questions: 'Wanjiru and Kamau, read us your new question.' Teacher responds to each new question by immediately connecting it to a future sub-strand or to the phenomenon: 'That is a brilliant question — the answer to that is going to come from Sub-Strand 1.7 Microorganisms.' Teacher photographs the final board (or assigns a learner to do so) and says: 'This photograph is your class's scientific record. In Lesson 8, we will return to this photograph and check how far we have come.'	Dot-Voting Epistemic Audit: The dot-sticker protocol is a low-stakes, high-visibility formative tool that makes collective epistemic confidence visible at a glance. The addition of new questions enacts the NGSS practice of 'Asking Questions' at a metacognitive level — learners demonstrate that deeper understanding generates richer questions, not fewer. Connecting the DQB to KEMRI's research practice contextualises the strategy in a Kenyan institutional frame.	Observable indicator: Every pair places at least one dot in each colour category AND contributes one new question card. Teacher checks that new question cards are genuine scientific questions (not statements). A concentration of RED dots on particular questions signals priority revision topics for Lesson 8. Teacher notes the ratio of green to red dots as a whole-class confidence indicator.

	Column C is your personal research gap list for Grade 10.'			
Model Building Phase 📹 **VIDEO:** [Analyzing a cumulative relative frequency graph] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Cumulative Frequencies (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners spend 25 minutes constructing or substantially updating their personal Cumulative Science Model — a mind-map or annotated diagram in their notebooks or on A4 paper. The model MUST include all four of the following required nodes, visually connected with labelled arrows: (1) The Anchoring Phenomenon — 'Mama Akinyi's Maize Farm, Kisumu: inferring a hidden soil pest'; (2) The Principle of Inference — definition plus one example from a Kenyan career; (3) At Least Two Kenyan Contexts — e.g., 'A doctor at Kenyatta National Hospital reads an X-ray,' 'A tea agronomist in Kericho tests soil pH,' 'A fisheries officer at Lake Victoria measures oxygen levels'; (4) At Least Two Science Topics from Lessons 1–6 — e.g., inference from indirect evidence, scientific processes, career mapping, data collection methods. Learners may add additional nodes freely. After 25 minutes, pairs swap models and conduct a structured peer-review using a provided KICD-aligned rubric checklist (four criteria: (a) all four required nodes present and labelled; (b) arrows are labelled with explanatory sentences; (c) Kenyan contexts are specific and accurate; (d) the model clearly shows how General Science connects to daily life). Reviewers write two 'stars' (strengths) and one 'wish' (one specific improvement) on a sticky note attached to their partner's model. Learners then make one final revision to their model based on	Before model building begins, teacher projects or draws on the board a sample 'skeleton model' with the four required nodes labelled but with all arrows and labels blank, saying: 'This is the minimum structure. Your model should go far beyond this skeleton — the richer your connections, the more powerful your model.' During the 25 minutes of model building, teacher circulates continuously — stop at 6–8 desks, spending no more than 90 seconds at each. Ask targeted questions: 'I see you have drawn the farmer — now draw an arrow to your career node and label it: what does YOUR career have in common with what she did?' WAIT TIME: 12 seconds. 'You have written 'doctor' — can you make it more specific? Which Kenyan hospital, and what kind of inference does that doctor make?' Before peer review begins, teacher models the star-wish protocol using a volunteer's model (with permission): 'My first star: the Kericho tea farm context is very specific and well-labelled. My wish: the arrow between inference and the anchoring phenomenon needs a sentence — what is the connection?' During the gallery walk, teacher narrates: 'Look for at least one career context you had never considered before — that is the power of learning together.' Close the lesson by returning to the driving question: 'How does a scientist know something is true when they cannot see it directly — and how is this power of General Science shaping life, careers, and the environment in Kenya? Look at	Cumulative Model Building with Structured Peer Review: Mind-map construction enacts the NGSS Science and Engineering Practice of 'Developing and Using Models.' The requirement that all four nodes be present and connected forces integration rather than isolated recall. The KICD-rubric-aligned peer review operationalises the values of Respect (considering peers' work seriously) and Responsibility (giving honest, constructive feedback). The gallery walk extends the community of knowledge beyond the individual, embodying the KICD core competency of Communication and Collaboration.	Observable indicators: (1) Each learner's model contains all four required nodes with labelled arrows (minimum standard). (2) Peer-review sticky notes contain two specific stars and one actionable wish — vague feedback ('good model') is redirected by teacher: 'Be specific — which node and which connection is strong, and why?' (3) At least one revision is made to each model after peer feedback. (4) During gallery walk, teacher counts models that receive 3 or more peer ticks as an indicator of models perceived as rich and Kenyan-contextualised. Teacher collects all models at lesson end for formative marking against KICD rubric descriptors before Lesson 8.

	peer feedback. The lesson closes with a 5-minute whole-class gallery walk: models are placed on desks and learners circulate, placing a small tick on any model that contains a Kenyan career context they had not thought of themselves.	your model. You have just answered that question in your own words, in your own career, in your own Kenya.'		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal: (1) DQB Curation — Which questions generated the most disagreement during sorting? What does this reveal about gaps in the evidence-gathering sequence from Lessons 1–6? Were there questions that no group could justify placing in Column A? If so, plan a brief 5-minute revisit at the start of Lesson 8. (2) Storyline Bridge — Did learners draw genuine connections between the principle of inference and their chosen careers, or were the career triangles superficial? If many triangles lacked specific Kenyan institutional contexts, consider providing a reference card for Lesson 8 listing 10 Kenyan science careers with their inferential practice (e.g., KEMRI epidemiologist, Kenya Meteorological Department forecaster, KEBS food safety analyst). (3) Model Building — Review the collected models before Lesson 8. Score each against the four KICD rubric criteria. Note: (a) How many models achieved all four required nodes? (b) How many Kenyan career contexts were unique across the class — compile a class 'Career Map of Kenya' to display in Lesson 8. (c) Were the arrow labels genuinely explanatory, or were they single-word labels? Single-word labels suggest learners are naming connections rather than understanding them — address this in the Lesson 8 showcase. (4) Peer Review Quality — Was the star-wish feedback specific and actionable? If most sticky notes were vague, model the protocol more explicitly at the start of Lesson 8. (5) Phenomenon Connection — Did every phase of today's lesson return to Mama Akinyi's maize farm in Kisumu? If the phenomenon connection felt forced or was dropped by learners, this suggests the phenomenon needs to be re-anchored more visually — consider displaying a photograph of a maize shamba from the Kisumu/Nyanza region prominently in the classroom for Lesson 8.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed the full Driving Question Board from Lessons 1–6 and sorted all question cards into three columns — 'We Can Now Answer This,' 'We Partly Understand This,' and 'We Still Don't Know This' — justifying each placement with specific evidence from previous lessons.
What did I learn?	Students learned that scientific understanding is cumulative and incomplete — some questions from their DQB can now be answered with evidence, others require more investigation in future sub-strands (The Cell, Nutrition, Respiration, etc.), and the principle of inference connects General Science to careers across Kenya, from doctors at Kenyatta National Hospital to fisheries officers at Lake Victoria to agronomists in Kericho.
How does this explain the phenomenon?	Students explained, through their Cumulative Science Models and peer-reviewed mind-maps, how the Kisumu maize farmer's act of inferring a hidden soil pest from indirect evidence is the same cognitive move used by scientists and professionals across Kenya — and how the entirety of General Science is built on this principle of making reliable conclusions from evidence that cannot always be directly observed.

LESSON 8 (40 minutes): Synthesis, Presentations and Unit Assessment — Final Explanation & Model Consolidation	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners synthesise the entire sub-strand by presenting career-inference charts, co-constructing a class-wide final answer to the Driving Question, completing a written formative assessment using a new Kenyan scenario, and comparing their initial mental models with their final models to mark growth in scientific understanding.
Knowledge	– Define and apply the principle of inference as a core scientific practice — distinguishing evidence, reasoning, and conclusion. – Explain how General Science is important in Kenyan daily life, environment, and technology across multiple domains. – Identify at least three Kenyan career pathways in which indirect evidence and inference are used professionally. – Articulate what General Science is as a discipline — its meaning, scope, and methods — using sub-strand vocabulary.
Skills	– Present a structured career-inference chart to peers with clarity and scientific accuracy. – Analyse a new Kenyan scenario (tea farmer in Kericho) to identify evidence, state a reasonable inference, and name a relevant science career. – Compare an initial (Lesson 1) mental model with a final (Lesson 8) revised model and describe what changed and why. – Construct and contribute to a class anchor-chart that answers the Driving Question with multiple lines of evidence.
Attitudes	– Appreciate that scientific knowledge is cumulative — each lesson's evidence added a layer to a growing, revised model. – Value the principle of inference as a tool that empowers Kenyans — farmers, doctors, environmental scientists — to act wisely on evidence they cannot see directly. – Demonstrate respect for peers' career aspirations and scientific reasoning during presentations and cold-calling.
Key Inquiry Question	How is General Science useful in daily life?
Purpose in Storyline	This is the culminating lesson of Sub-Strand 1.1. Learners bring all seven lessons of evidence and reasoning to bear on the Driving Question — How does a scientist know something is true when they cannot see it directly? — producing a final class-constructed answer grounded in the Kisumu maize-farmer phenomenon and extended to a new Kenyan scenario. The lesson closes the DQB by marking answered questions and flagging unanswered questions as entry points for Sub-Strand 1.2: The Cell.
Safety Notes	No laboratory hazards. Ensure equitable airtime during presentations — use a timer visible to all groups. If the class is large, cold-call contributions to avoid dominance by a single group.

B. LESSON OVERVIEW

Lesson 8 is the capstone of Sub-Strand 1.1: Introduction to General Science. After seven lessons of building evidence — from the Kisumu maize-farmer phenomenon, through the meaning and importance of General Science, career exploration, the formal principle of inference, the Mystery Bags hands-on activity, and the Driving Question Board (DQB) curation — learners now synthesise everything into a final, public, evidence-based answer to the unit Driving Question: "How does a scientist know something is true when they cannot see it directly — and how is this power of General Science shaping life, careers, and the environment in Kenya?"

The lesson moves through five structured phases. In the Predict Phase, learners make a final individual prediction of the Driving Question answer before seeing their peers' work — activating metacognitive awareness of how their thinking has changed since Lesson 1. In the Observe Phase, each group presents its career-inference chart (2 minutes per group) while the class records connections on a structured listening frame. In the Explain Phase, the class co-constructs a final anchor-chart answer to the Driving Question using cold-called contributions, and then each learner independently completes a short written formative task — a new Kenyan scenario involving a tea farmer in Kericho noticing unusual leaf curl — in which they must identify evidence, state a reasonable inference, and name a General Science career that would investigate it professionally. The DQB Phase closes the board: answered questions are marked with a green tick; remaining questions (which require knowledge of the cell, nutrition, and respiration) are flagged as preview questions for Sub-Strand 1.2. The Model Building Phase asks every learner to compare their Lesson 1 sketch model with their final revised model, annotate what changed and why, and share one sentence of growth with the class.

The lesson ends with a forward-looking transition: the teacher uses the Kericho tea-farmer scenario to bridge into Sub-Strand 1.2 — "To understand exactly how the maize root is being destroyed underground — or why the tea leaf is curling in Kericho — we first need to understand what a cell is. That is where we go next." This ensures the phenomenon does not close but opens into the next sub-strand, maintaining the CBE storyline thread across the year.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** When to use z or t statistics in significance tests Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Scientific Notation Values (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner opens their notebook to a fresh page and writes a solo response — in 3–5 sentences — to this prompt: 'Before you see your classmates' presentations, write your best current answer to the Driving Question: How does a scientist know something is true when they cannot see it directly? Use the maize farmer in Kisumu in your answer.' They then write one sentence completing: 'In Lesson 1, I thought… but now I think…' to activate metacognitive comparison. No peer discussion yet — this is an individual written commitment.	Write the Driving Question in full on the board — leave it there for the entire lesson. Say: 'Before anyone shares, I want to know what YOU think right now — after everything we have done together. Open your notebook. Three to five sentences. Use the maize farmer. Go.' Allow 4 minutes of silent writing. Circulate and scan notebooks — do NOT comment yet. After 4 minutes, say: 'Now write one more sentence — starting with: In Lesson 1, I thought… but now I think…' Allow 90 seconds. Cold-call 2 learners to read their 'I thought / I think' sentence aloud before moving on. WAIT TIME: 10 seconds after each cold-call before accepting the answer. Affirm both responses without evaluating correctness — say: 'Hold that thought — your presentations are about to test it.'	Metacognitive Bookend — learners articulate their current best understanding BEFORE seeing peers' work, creating a personal benchmark they will compare against at the Model Building Phase. This strategy surfaces prior reasoning without anchoring bias from group presentations, and it gives the teacher a real-time window into individual understanding before summative evidence is gathered.	Teacher scans notebooks during the 4-minute writing window. Look for: (1) presence of the word 'evidence' or 'inference' — indicates uptake of Lesson 5 vocabulary; (2) reference to the maize farmer as an example — indicates phenomenon connection; (3) generalisation beyond the farmer to other careers — indicates transfer. Flag 2–3 notebooks showing weak phenomenon connection for targeted cold-calling during the Explain Phase.
Observe Phase 📹 **VIDEO:** When to use z or t statistics in significance tests Source: Khan	Each group presents its career-inference chart for exactly 2 minutes. The presenting group must answer: 'Name your chosen career, explain one task in that career that uses inference, and connect it back to the Kisumu	Before presentations begin, distribute or display the 2-minute timer visibly. Say: 'Every group will present for exactly two minutes. Your job as listeners is to fill in your listening frame — Career, Evidence, Inference. I will cold-call	Structured Listening Frame — rather than passive watching, learners actively categorise each presentation into evidence/inference columns, turning peer presentations into a second round of data collection.	Circulate during presentations to check listening frames. Look for: (1) correct use of 'evidence' vs 'inference' in the columns — misuse indicates a persistent conflation that needs re-teaching in the Explain Phase; (2) whether

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Scientific Notation Values (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	maize farmer.' While each group presents, all other learners complete a structured listening frame in their notebooks with three columns: Career Named | Evidence Used in That Career | How Inference Appears. After all presentations, learners have 90 seconds to review their listening frame and star the career they found most surprising or most connected to the phenomenon.	listeners after each presentation, so stay sharp.' Assign presentation order (e.g., by row). For each group, start the timer and say: 'Group [name] — go.' At the 1:45 mark, give a quiet hand signal: 'Thirty seconds.' After each presentation, cold-call ONE listener (not from the presenting group): 'Tell me one piece of evidence this career uses that we did NOT observe directly.' WAIT TIME: 10 seconds. After all groups have presented, say: 'Star the career in your frame that surprised you most. You have 90 seconds.' Cold-call 2 starred responses before moving to the Explain Phase.	The 'surprising career' star creates a low-stakes personal evaluation that primes the whole-class synthesis in the Explain Phase.	learners are connecting each career back to the phenomenon or treating careers as isolated facts. Ask any learner who appears disengaged: 'What column are you writing in right now?' as a low-stakes accountability check.
Explain Phase 📹 **VIDEO:** When to use z or t statistics in significance tests Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Scientific Notation Values (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Part A — Class Anchor Chart (8 minutes): The class co-constructs a large anchor-chart answer to the Driving Question. Each cold-called learner contributes one clause to a growing class sentence. The final anchor-chart sentence must include: (a) what the scientist does, (b) what kind of evidence they use, (c) the word 'inference', and (d) a Kenyan example. Part B — Written Formative Task (8 minutes, individual and silent): Learners receive or copy this new scenario: 'A tea farmer in Kericho notices that the leaves on several tea bushes along one row are curling inward and turning pale, even though rainfall and fertiliser application have been the same across the entire farm. She has not seen any insect or disease on the leaves.' Learners must answer three questions in their notebooks: (1) List TWO pieces of evidence the farmer has. (2) State ONE reasonable inference about what might be causing the problem. (3)	Part A: Draw a large box on the board headed: 'OUR FINAL ANSWER TO THE DRIVING QUESTION.' Say: 'We are going to build one class sentence together. I will cold-call six people. Each person adds one clause. The sentence must include: what the scientist does, the kind of evidence they use, the word inference, and a Kenyan example.' Cold-call 6 learners in sequence — WAIT TIME 10 seconds each — and write each clause on the board as it is given. After the sixth clause, read the full sentence aloud and say: 'Does everyone agree this answers our Driving Question? Thumbs up, sideways, or down.' Adjust any clause the class thumbs-down before writing the final version on the anchor chart. Part B: Say: 'Now I want to see what YOU can do alone. New scenario — Kericho. Three questions. Eight minutes. Silent.' Read the scenario aloud once slowly. Write the three question	Collaborative Sentence Building (Anchor Chart) followed by Independent Transfer Task — the two-part structure is deliberate: collaborative co-construction externalises the class's shared understanding and models the scientific practice of building consensus from evidence; the independent transfer task then reveals whether each individual learner can apply the same reasoning to a new but structurally identical Kenyan scenario. The Kericho tea scenario is isomorphic to the Kisumu maize scenario — same logical structure (hidden cause, visible symptoms, indirect evidence, inference) — making transfer both challenging and achievable.	Collect written formative tasks (or photograph them). Grade against a 3-point rubric: (1) Evidence — did the learner name two observable facts from the scenario, not inferences? (0–1 marks) (2) Inference — is the stated inference a logical conclusion about a hidden cause, not a restatement of the symptom? (0–1 marks) (3) Career + indirect evidence — does the learner name a real General Science career AND explain a method of indirect evidence collection relevant to that career? (0–1 marks). Learners scoring 0–1 are flagged for individual follow-up. A learner who writes 'the inference is that the leaves are curling' (restating observation) has not yet grasped the observation/inference distinction — this is the critical misconception to address before Sub-Strand 1.2.

	Name ONE General Science career whose professional would investigate this — and explain in one sentence how they would use indirect evidence.	stems on the board: (1) TWO pieces of evidence. (2) ONE inference. (3) ONE career + how they use indirect evidence. Circulate silently — do not answer questions. At 6 minutes, say quietly: 'Two minutes remaining — check question 3.' After 8 minutes, collect written responses OR ask learners to fold and keep for return next lesson.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** When to use z or t statistics in significance tests Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Scientific Notation Values (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners approach the DQB (physical board or digital equivalent) as a class. Working in silence for 2 minutes, they re-read all the sticky notes or cards added across Lessons 1–7. The teacher then facilitates a 4-minute whole-class ceremony: (1) Green tick — any question the class can now answer with evidence from this sub-strand. (2) Yellow arrow — any question that is partially answered but needs more investigation. (3) Red question mark — any question that is explicitly unanswered and will be addressed in a future sub-strand. Each learner then writes one personal sentence in their notebook: 'The question I most want to answer next is _______ because _______.'	Say: 'Our Driving Question Board has been with us since Lesson 1. Let us honour it properly.' Walk to the board. Point to the first sticky note. Say: 'Read this question. Can we answer it now — yes, partly, or not yet? Hands — full answer, show five fingers; partial, show three; not yet, show one.' Tally quickly and assign the colour code (green tick, yellow arrow, red question mark) with the class watching. Repeat for 4–5 key DQB questions — prioritise: (a) 'What is General Science?' (should be fully answered — green tick), (b) 'How does a scientist know something is true?' (fully answered — green tick), (c) 'What is inside a maize root?' (not yet answered — red question mark). Point to the red-question-mark cards and say: 'These are our gifts to future lessons. Write in your notebook: The question I most want to answer next is _______ because _______.' WAIT TIME: allow 60 seconds of writing. Cold-call 2 learners to share. Say: 'Hold those questions — Sub-Strand 1.2, The Cell, will begin answering them.'	DQB Closure Ceremony with Colour-Coded Status Markers — this strategy transforms the DQB from a display into a living record of growing understanding. Assigning green ticks, yellow arrows, and red question marks makes the epistemological structure of science visible: science never closes all questions — it answers some and opens more. The personal 'question I most want to answer next' sentence builds intrinsic motivation and provides the teacher with a formative window into each learner's curiosity profile.	Listen for the quality of justification learners give when voting on whether a DQB question is 'fully answered'. A learner who says 'we can answer it because inference means using evidence' is demonstrating conceptual understanding. A learner who says 'we did that in Lesson 5' is citing lesson memory without conceptual grasp — prompt: 'Can you say what inference means in your own words?' Note whether the 'question I most want to answer next' sentences show phenomenon connection (e.g., 'I want to know what is actually destroying the maize roots') or generic curiosity (e.g., 'I want to know about science') — the former indicates deeper anchoring to the storyline.
Model Building Phase 📹 **VIDEO:**	Learners open their notebooks to their Lesson 1 sketch model — the drawing or diagram they made at the very beginning of the unit	Say: 'Find your Lesson 1 model. Look at it for 30 seconds. Do not judge it — that was your best thinking on Day 1.' Allow 30	Model Comparison and Annotation (L1 vs L8) — placing the initial and final models side by side makes conceptual growth visible and	Walk the room during the 5-minute drawing window. Look for: (1) Is the inference step clearly labelled — or is the learner drawing only

When to use z or t statistics in significance tests Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Scientific Notation Values (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	showing 'how the maize farmer knows something is true even though she cannot see it directly.' Beside it, they draw or annotate their final model — which must now include: (a) the farmer/scientist figure, (b) labelled arrows showing 'Direct Evidence' and 'Indirect Evidence', (c) the inference step labelled with the word 'INFERENCE', (d) at least one Kenyan career role (e.g., KALRO agronomist, KNH lab technologist, Kericho Tea Research Institute scientist), and (e) one arrow pointing to 'Sub-Strand 1.2: The Cell — unanswered questions'. Learners write two sentences below their final model: 'My model changed because… / My model shows that General Science helps Kenya by…' The lesson closes with the teacher reading the Sub-Strand 1.2 bridge statement aloud.	seconds of quiet looking. Then say: 'Now, right next to it, draw your final model. It must show five things —' (write on board): '1. The farmer or scientist. 2. Direct and indirect evidence arrows. 3. The word INFERENCE. 4. One Kenyan career. 5. An arrow pointing to Sub-Strand 1.2 questions.' Allow 5 minutes of drawing. Then say: 'Write two sentences below your model — starting with: My model changed because… and My model shows that General Science helps Kenya by…' Allow 2 minutes of writing. Cold-call 3 learners to share one sentence each. WAIT TIME: 10 seconds per cold-call. After the third share, stand at the board and say: 'To understand how the maize root is being destroyed — or why the tea leaf is curling in Kericho — we first need to understand what a cell is. Every plant, every root, every leaf is made of cells. That is where we go next: Sub-Strand 1.2, The Cell.' Point to the red-question-mark cards on the DQB and say: 'Those questions are waiting for us.'	concrete, not abstract. The mandatory five-element checklist for the final model ensures completeness without over-scaffolding. The two-sentence written reflection ('My model changed because… / My model shows that…') requires learners to articulate change, not merely display it — this is a higher-order metacognitive demand consistent with CBE's emphasis on self-efficacy and learning to learn. The bridge statement connects the completed model to the next sub-strand, maintaining the phenomenon's explanatory power beyond Sub-Strand 1.1.	the evidence without showing the reasoning step? (2) Is the Kenyan career grounded in indirect evidence, or is it listed without connection? (3) Does the arrow to Sub-Strand 1.2 show any specific question (e.g., 'what is inside the root?') or is it blank? Cold-call the 3 learners whose model comparison sentences are most substantive — these become model exemplars for peer display. For learners whose final model is nearly identical to their Lesson 1 model, ask privately: 'What is one thing you know now that you did not know in Lesson 1?' — this probes whether absence of model change reflects genuine non-learning or drawing difficulty.

D. TEACHER REFLECTION

1. During the group presentations, did every group correctly connect their chosen career to the principle of inference — or did some groups describe career tasks without explaining how indirect evidence is used? What would I do differently to scaffold the inference-connection requirement before the next time learners prepare their charts?

2. Looking at the written formative tasks from the Kericho tea-farmer scenario: how many learners correctly distinguished between an observation and an inference? If more than one-third conflated the two, what targeted re-teaching strategy will I use at the start of Sub-Strand 1.2 before introducing the cell?

3. Did the DQB closure ceremony feel rushed, or did learners genuinely engage with the green/yellow/red status decisions? Were learners citing evidence for their colour choices, or were they simply voting by memory? How can I build stronger evidence-citation habits into future DQB sessions?

4. Comparing Lesson 1 and Lesson 8 models: did the majority of learners show visible conceptual growth — specifically, did they add the inference step as a distinct labelled element that was absent in Lesson 1? Which learners showed the least growth, and is this a conceptual gap, a language barrier, or a drawing-confidence issue?

5. When I delivered the Sub-Strand 1.2 bridge statement ('To understand how the maize root is being destroyed, we first need to understand what a cell is'), did learners respond with genuine curiosity — questions, leaning forward, murmurs? If not, how can I make the bridge feel like an unresolved mystery rather than a textbook transition, so that the phenomenon continues to drive motivation into the next sub-strand?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed all peer groups present their career-inference charts, recording in a structured listening frame how each career — from KALRO agronomists to KNH lab technologists to Kericho Tea Research Institute scientists — uses indirect evidence and inference as a core professional practice. They observed the class Driving Question Board being colour-coded for the first time: green ticks for questions fully answered, yellow arrows for partial answers, and red question marks for questions (especially about the cell and root destruction) that remain open and will anchor Sub-Strand 1.2.
What did I learn?	Learners learned that the principle of inference — drawing a reliable conclusion about something not directly observable, using evidence and prior knowledge — is not a single lesson topic but the unifying thread of the entire sub-strand and of General Science as a discipline. They learned that a new but structurally identical Kenyan scenario (the Kericho tea farmer's curling leaves) can be analysed using the same three-part framework: identify evidence, state an inference about a hidden cause, and name a career that would investigate it professionally. They also learned that their understanding has grown measurably since Lesson 1 — visible in the annotated comparison of their initial and final models.
How does this explain the phenomenon?	The final anchor-chart answer co-constructed by the class directly explains the anchoring phenomenon: the Kisumu maize farmer knew a soil pest was destroying her roots without seeing it because she applied the principle of inference — she identified a pattern of indirect evidence (localised wilting, yellowing, absence of other causes, season of year), used her prior knowledge of maize pests, and drew a reliable conclusion about a hidden cause. This is not unique to farmers — it is the daily practice of every General Science professional in Kenya, and it is the foundation of everything the class will study in Sub-Strands 1.2 through 3.4.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.